# Welfare Reform and the Earned Income Tax Credit in the 1990s*

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**Abstract**

A large literature credits the 1993 Earned Income Tax Credit (EITC) expansion with increasing labor supply among single mothers in the 1990s, using difference-in-differences designs that compare mothers with different numbers of children. I show that, once employment trends are allowed to vary with pre-reform welfare exposure, the estimated effect of the EITC falls to zero in replications of four representative studies. The original designs violate the parallel trends assumption because treatment and control groups differ systematically in exposure to welfare policy changes. Placebo tests and a reweighting exercise show that the canonical estimates reflect compositional imbalance between the treatment and control groups. A simple machine-learning procedure would have flagged this violation ex ante. The results suggest that the difference-in-differences estimates are driven by differential exposure to welfare reform rather than by differences in EITC generosity.

**Keywords:** Earned Income Tax Credit (EITC), Labor Supply, Welfare Reform, Difference-in-Differences
**JEL Classification:** I38, H24, J22

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# 1 Introduction

A large body of research credits the 1993 expansion of the Earned Income Tax Credit (EITC) with the large and persistent increase in labor supply and the decline in welfare use among single mothers in the 1990s. In fact, the 1993 expansion is particularly important because it underpins much of what we know about the EITC's labor supply effects [Nichols and Rothstein, 2016] and is also the basis for estimates of its effects on broader outcomes—including earnings, income, poverty, health, and human capital. In part because the 1990s changes are viewed as driven by the EITC's work-contingent subsidies, they contributed to a shift in social policy after which "virtually all gains in spending on the social safety net for children since 1990 have gone to families with earnings" [Hoynes and Schanzenbach, 2018]. The 1990s experience continues to shape today's debates over work-contingent tax and transfer policy.

A longstanding concern is that the estimated effects of the 1993 EITC are confounded by the contemporaneous changes to cash welfare—"welfare reform"—implemented through federal law and earlier state waivers [e.g., Mead, 2014, Kleven, 2024]. In particular, ethnographic histories suggest that administrative barriers and changing norms associated with welfare reform were central drivers of declining welfare caseloads and rising employment [DeParle, 2004, Weaver, 2000]. Standard difference-in-differences (DiD) estimates of the effect of the EITC address this concern by including state-by-year fixed effects or similar controls for state-level policies and economic conditions. The concern, however, is that welfare reform affected individuals in proportion to their reliance on the welfare system. If so, the relevant variation is within states rather than across them, and state-by-year controls are insufficient to eliminate bias.

In this paper, I show that the identifying assumptions underlying canonical DiD designs comparing women with different numbers of children are violated because the groups differ systematically in exposure to welfare reform, and that difference drives the estimated effects. When DiD specifications are augmented to include controls for the interaction of year dummies with pre-reform welfare exposure—time shocks proportional to welfare use—the estimated effect of the EITC is attenuated to zero in all of the studies, and these time effects account for the canonical treatment–control contrast in employment growth.

While this pattern suggests that the employment divergence attributed to the EITC instead reflects differential exposure to welfare reform, a natural concern is that controlling for welfare exposure also absorbs genuine EITC variation, since EITC treatment and welfare exposure are correlated. To formalize the confounding mechanism and address this concern, I propose two tests. First, I redraw the standard time series graphs illustrating the differences between EITC treatment and control groups over time after reweighting the sample to equalize predicted pre-reform welfare exposure between groups. This shows that the relative changes in employment that identify the DiD estimate of the EITC arise entirely from differences in the composition of the treatment and control groups. Second, I estimate "placebo" DiD regressions that compare individuals facing the same EITC schedule, and show these specifications generate large, spurious "effects" that are directly proportional to the difference in predicted pre-reform welfare exposure between compared groups. I show these placebo effects are evidence of omitted variables bias, and the source of the bias is the exclusion of controls for exposure to welfare reform.

From the perspective of empirical practice, it is undesirable that this source of bias remained undetected for so long, despite the omitted variable being available in the data. To address this problem prospectively, I propose using a simple machine-learning (ML) algorithm for covariate selection applied to the standard graphical analysis used to assess the "parallel trends" assumption in DiD estimators. Specifically, I ask the ML model to select covariates that predict annual employment within one group (e.g., the control group) and use those covariates to out-of-sample forecast outcomes in the other group (e.g., the treatment group). This provides a prediction of the counterfactual outcome that is constructed to exclude the effect of the difference in treatment. Using the example of the EITC, this procedure selects time effects in pre-reform welfare exposure as important covariates, and the resulting predicted counterfactual outcomes closely match realized outcomes, leaving no room for an EITC-driven treatment effect. In short, this visualization would have caused researchers to reject the parallel trends assumption ex ante—before any treatment-effect estimation.

This paper makes three contributions. First, it shows that the canonical difference-in-differences designs used to evaluate the 1993 EITC expansion do not identify the effect of the EITC under their maintained parallel-trends assumption. Falsification tests show the estimator reproduces the published "effects" with no treated-versus-untreated contrast at all; a formal decomposition gives the mechanical source of the bias; reweighting on pre-reform welfare exposure eliminates the treatment-control divergence; and the same confound affects an unrelated identification strategy based on welfare time limits. Second, it provides a direct robustness diagnostic for these specifications. Adding Year × predicted pre-reform welfare exposure tests whether the estimated EITC effects survive once employment trends are allowed to vary flexibly with baseline exposure to the welfare system. In each of four influential specifications, the estimated EITC effects collapse to zero. Third, it contributes a general design diagnostic. A simple machine-learning cross-prediction procedure provides a transparent graphical test of parallel trends that would have flagged this violation ex ante and can be applied prospectively in other difference-in-differences settings. Taken together, the evidence suggests that the employment divergence attributed to the EITC in the canonical designs was generated by differential exposure to welfare reform, not by differences in EITC generosity.

The rest of the paper proceeds as follows. Section 2 provides background and introduces the proportional-exposure hypothesis, with descriptive evidence motivating the reanalysis. Section 3 reports replications of canonical EITC specifications, showing that controlling for time × pre-reform welfare exposure eliminates the estimated effects, and links to replication details in Appendix B. Section 4 explains why the bias arises, using a decomposition of the DiD estimator, reweighting, and falsification tests that demonstrate the spurious nature of the canonical estimates. Section 5 uses machine-learning cross-predictions as a design diagnostic to surface the omitted confounder ex ante. Section 6 concludes with implications for research that relies on the 1993 expansion (poverty, child outcomes, maternal and infant health) and for current policy debates.

## 2 Background and Literature Review

An extensive literature credits the 1993 Earned Income Tax Credit (EITC) expansion with historic increases in employment and reductions in welfare receipt among single mothers in the 1990s [see reviews in Blank, 2002, Hotz and Scholz, 2001, Nichols and Rothstein, 2016]. These conclusions are based on difference-in-differences (DiD) designs comparing more-treated groups (e.g., mothers of two or more children) to less-treated or untreated groups (mothers of one child or single women without children), and attributing the post-1993 divergence in employment trends to the EITC. The classic starting point is Eissa and Liebman [1996], who study the 1986 EITC expansion by comparing labor-supply changes among single women with children to those among single women without children. The 1986 expansion preceded the sweeping welfare changes of the mid-1990s.

The 1993 expansion was the largest in the program's history: the maximum credit for families with two or more children more than doubled, from \$1,511 in 1993 to \$3,556 when fully phased in by 1996, the phase-in rate rose from 19.5 to 40 percent, and a small credit was extended to childless workers for the first time. A typical estimating equation has the following form:

$$y_{ist} = \theta_{st} + \beta\,\mathrm{EITC}_{ist} + \gamma\,\mathrm{Children}_i + X_{ist}\delta + \epsilon_{ist}, \tag{1}$$

where $y_{ist}$ is an indicator for employment of individual $i$ in state $s$ and year $t$, and $\theta_{st}$ are state $\times$ year fixed effects. $\mathrm{EITC}_{ist}$ captures variation in EITC generosity, and $\mathrm{Children}_i$ are dummies for the number of children. The vector $X_{ist}$ includes standard demographic controls such as education, race, marital status, age, and age of youngest child.

The identifying assumption of this model is that, absent changes in EITC policy, outcomes for individuals with different numbers of children within a given state would have followed parallel trends over time, conditional on state $\times$ year effects and the covariates $X_{ist}$. However, the mid-1990s also saw sweeping welfare reforms—both through federal legislation [U.S. Congress, 1996] and earlier state "waivers" allowing states to implement pro-work changes in program rules and sanctions for non-compliance—that dramatically reduced caseloads [for a comprehensive review, see Ziliak, 2016]. Ethnographic accounts suggest that these changes and associated administrative barriers to welfare receipt were the major causes of declining welfare use and rising employment rather than employment subsidies like the EITC. The Rockefeller Institute's implementation study, for example, found that many local sites had reoriented their efforts toward diverting would-be recipients away from welfare and toward work. "Local officials and workers view state officials as wanting to see, above all else, lower [welfare] caseloads" (as quoted in Weaver 2000). In areas with the most dramatic reductions in welfare use, like Wisconsin, ethnographers attributed the changes to work requirements enforced by new bureaucratic procedures [DeParle, 2004]. As one observer summarized, "the hassle factor of welfare is much more powerful in pushing them off the rolls (and consequently into jobs) than the vaguer promise of later wage subsidies" [as quoted in Mead, 2014].

These ethnographic and implementation accounts suggest two features of welfare reform that matter for identification. First, welfare reform is unlikely to be summarized exclusively by the changes enumerated in legislation and the formal dates on which states enacted waivers or TANF rules. Second, because these changes operated through the welfare system itself, their effects should have been largest for groups most exposed to that system before reform. Groups with higher pre-reform welfare exposure—such as mothers of two or more children—were therefore more exposed to changes in welfare administration and enforcement, regardless of EITC eligibility.

Figure 1 illustrates this concern. It presents a binned scatterplot of the change in employment of single women between the pre-reform (1991–1993) and post-reform (1999–2001) periods, plotted against each group's predicted pre-reform welfare exposure.[1]

The figure shows a strong positive relationship: groups with higher predicted welfare exposure experienced larger subsequent increases in employment. The slope of the regression line is 0.65, implying that a ten-percentage-point increase in predicted pre-reform welfare exposure for any group is associated with a 6.5 percentage-point increase in employment.

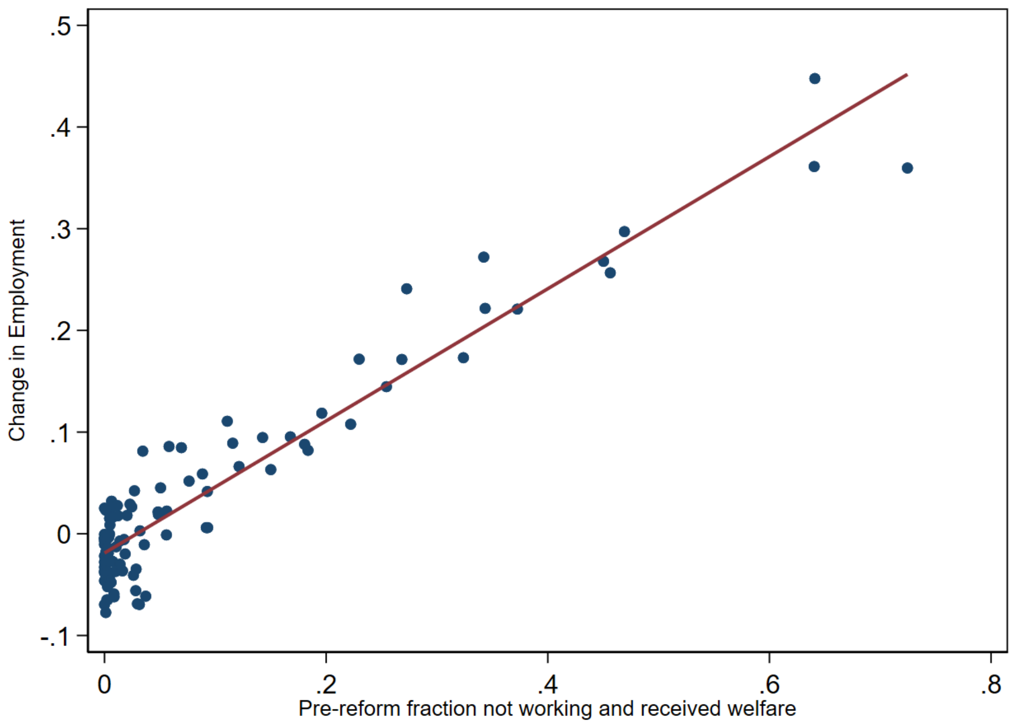
A back-of-the-envelope calculation illustrates the implications. In the pre-reform period, about 12.8 percent of single mothers with one child were on welfare and not working, compared to 26.4 percent of mothers with two or more children. Applying the slope in Figure 1 to these baseline exposure rates predicts employment gains of roughly 8 and 17 percentage points, respectively; the actual changes were 7 and 17. The fitted intercept of −0.02 shifts both predictions equally and drops out of their difference. The canonical DiD estimate is thus fully predicted by pre-reform welfare exposure alone—without invoking any EITC effect.

Appendix F expands this calculation into a Monte Carlo exercise. Starting from a single pre-reform cross section, I simulate a policy that randomly removes mothers from welfare and assigns non-working mothers to employment at rates matching the actual annual changes observed in the 1990s, with no differential treatment by number of children, education, or any other characteristic. This simple counterfactual policy reproduces the canonical DiD estimates almost exactly and replicates "puzzles" like the fanning-out of employment changes by family size and age of youngest child highlighted by Kleven [2024].

This paper builds on the closely related work of Kleven [2024], which identifies welfare reform as the leading alternative explanation for the employment changes attributed to the 1993 EITC expansion. Kleven shows that the estimated effects of the expansion increase with welfare-treatment intensity, proxied by the age of the youngest child or by predicted AFDC receipt. That pattern is highly suggestive, but not by itself dispositive: it could

[1] Welfare exposure for individual $i$ is defined as the predicted probability of being on welfare and not working in 1993: $W_i \equiv \Pr(wno_i = 1 \mid X_i) = \Phi(X_i\alpha)$, where $X_i$ includes indicators for education, race, marital status, state of residence, age of youngest child, and a flexible polynomial in age with interactions by education. The model is estimated separately by number of children $(0, 1, 2+)$ using data from 1991 to 1993. Because the predictors and the estimation sample use only pre-reform information, $W_i$ is predetermined with respect to the post-reform employment changes analyzed below.

Figure 1: Employment Changes and Pre-Reform Welfare Exposure




Notes: Each point represents a percentile bin of single women grouped by predicted pre-reform probability of receiving welfare and not working. The horizontal axis shows each bin's pre-reform share on welfare and not working; the vertical axis shows the change in the bin's employment rate between the pre-reform (1991–1993) and post-reform (1999–2001) period averages. The fitted line has slope 0.65 (s.e. 0.02) and intercept −0.02. Sample: Single women ages 20–50 from March CPS 1991–1993 and 1999–2001 (Flood et al. [2024]).

reflect confounding from welfare reform, or it could reflect genuine EITC responses that are larger among mothers with greater attachment to the welfare system. Nor did Kleven's evidence fully resolve objections based on the timing and intensity of formal state welfare-policy changes. The question left open is therefore whether welfare exposure operates as an omitted variable in the canonical designs, and whether that omitted variable is large enough to account for the published estimates.

The absence of a quantitative resolution has allowed disagreement in the literature to persist. For example, Schanzenbach and Strain [2021] reject Kleven's critique (responding to its 2019 working-paper version), arguing that the additional controls in his empirical strategy absorb the true effect of the EITC. With specific regard to the 1993 expansion, they emphasize that state AFDC waivers created heterogeneity in the timing and intensity of welfare reform, and contend that the EITC expansion increased employment even in states that had not yet implemented waiver policies. On this basis, they conclude that the expansion raised employment independently of welfare reform and that the canonical interpretation of the 1990s evidence remains valid. Appendix D shows, however, that the relationship between pre-reform welfare exposure and subsequent employment growth is nearly identical in waiver and non-waiver states (slopes 0.62 and 0.67), suggesting that the relevant exposure gradient is not driven solely by formal waiver timing but reflects broader nationwide changes associated with welfare reform.

Meanwhile, subsequent published work (e.g., Michelmore and Pilkauskas 2021) continues to rely on the same canonical DiD design without directly addressing this identification issue.

The central question, therefore, is whether differential exposure to welfare reform biases the canonical difference-in-differences estimator. Despite the prominence of this debate, the literature offers little econometric guidance on how to adjudicate it. The next sections propose and implement several empirical tests that establish the presence of bias, quantify its magnitude, and provide a corrective specification.

# 3 Replication Controlling for Welfare Exposure

I begin by showing that directly controlling for pre-reform welfare exposure eliminates the estimated effect of the EITC in the specifications used by four influential papers that examine the 1993 expansion: Schanzenbach and Strain [2021], Meyer and Rosenbaum [2001], Hoynes and Patel [2018], and Michelmore and Pilkauskas [2021]. These studies are representative of the published evidence on the EITC, use widely known data from the March CPS, but differ in their parameterization of the EITC, the subpopulations studied, and whether they incorporate only federal or both federal and state expansions.

I focus on the period 1991–2001, which brackets the 1993 expansion and the 1996 federal welfare reform while excluding earlier and later policy changes.[2] Following the literature, I restrict the sample to unmarried women ages 20–50, dropping mothers whose youngest child is older than 18 and retaining childless women as a comparison group, and I include covariates standard in the literature: a quartic in age, dummies for education, race, age of youngest child, and marital status. All specifications include state-by-year fixed effects, and the basic estimating framework follows Equation 1.[3] The dependent variable is an indicator for employment last year (positive earnings), although similar results hold for alternative employment and earnings measures.

My replication (documented in Appendix B) closely mirrors the published results, and each specification produces economically large and statistically significant estimates of the 1993 EITC expansion's effect on employment.

**Augmenting with pre-reform welfare exposure.** I next show what happens when each specification is augmented with an interaction of year dummies and the individual's pre-reform welfare exposure rate.[4] This control captures the possibility that welfare reform

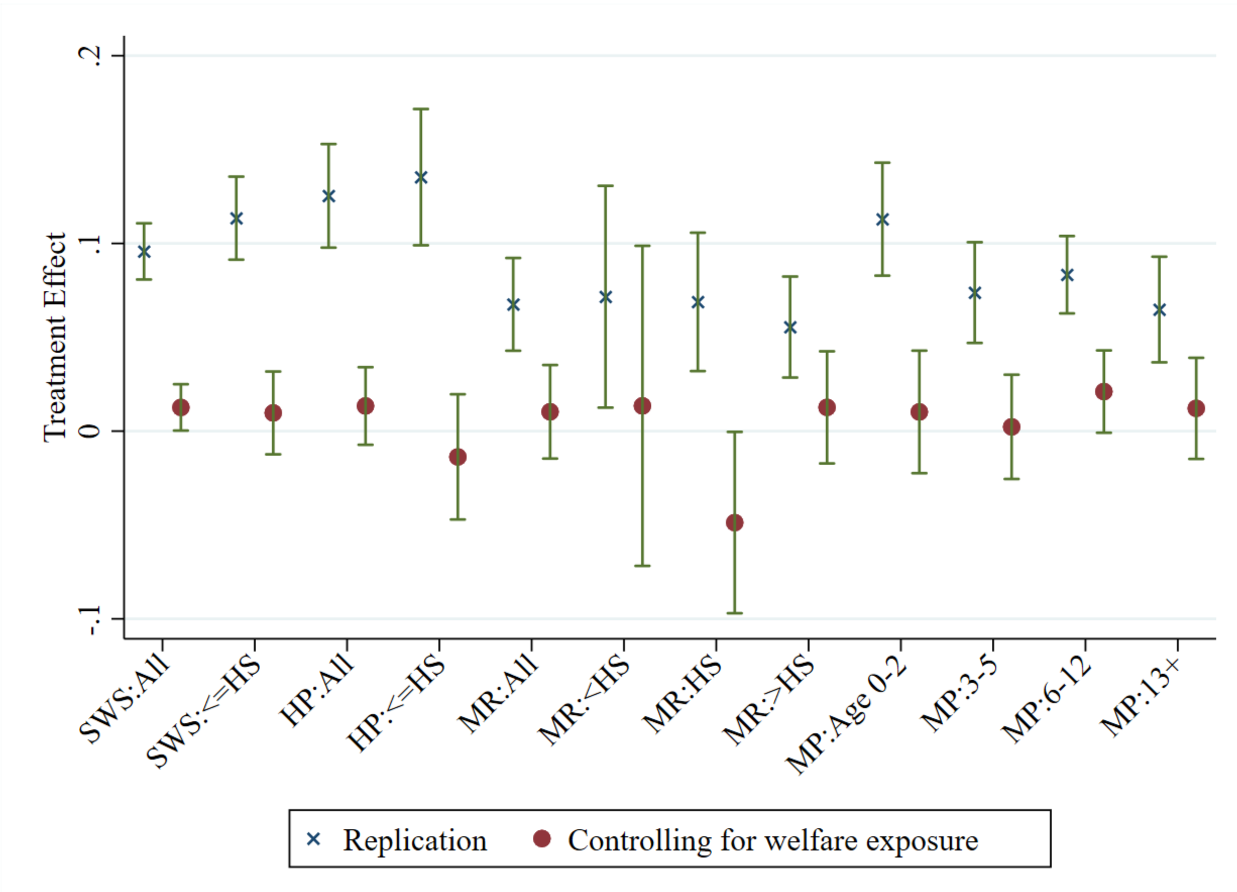
[2]Meyer and Rosenbaum [2001] analyze 1984–1996, Hoynes and Patel [2018] study 1991–1998, Schanzenbach and Strain [2021] examine 1989–1998, and Michelmore and Pilkauskas [2021] study 1990–2016. I replicate the results of these studies over a consistent 1991–2001 period except for Meyer and Rosenbaum [2001], where I analyze only 1991–1996.

[3]Summary statistics for this sample are provided in Appendix A.

[4]This rate is the same measure of predicted welfare exposure introduced above in connection with Figure 1.

effects were proportional to prior welfare use and therefore differentially affected treatment and control groups.

Figure 2: Replication of Canonical EITC Specifications with and without Controlling for Time × Pre-Reform Welfare Use




Notes: "×" markers show baseline replication estimates; circles show estimates after adding Year × predicted pre-reform welfare exposure interactions. Vertical bars are 95% confidence intervals clustered at the state-by-year level. SWS = Schanzenbach and Strain [2021]; HP = Hoynes and Patel [2018]; MR = Meyer and Rosenbaum [2001]; MP = Michelmore and Pilkauskas [2021]. Sample subgroups: All = all single women; ≤HS = high school or less; <HS = less than high school; >HS = more than high school. MP specifications interact EITC with age of youngest child (0–2, 3–5, 6–12, 13+). Sample: Single women ages 20–50, March CPS 1991–2001.

The results are shown in Figure 2. In every case, including this interaction term causes the EITC effect estimates to collapse toward zero and become statistically insignificant. This pattern holds across all four specifications, despite differences in EITC parameterization, sample restrictions, and treatment variables.

These findings are consistent with the view that the differential change in employment between treatment and control groups in the 1990s is fully explained by their differential exposure to welfare reform, not by their differential EITC treatment intensity. Since $W_i$ is a pre-treatment, time-invariant characteristic estimated only with 1991–1993 data and is not conditioned on post-reform behavior or contemporaneous policy take-up, interacting it with year dummies simply allows differential time paths proportional to baseline exposure, which is precisely the potential bias the specification seeks to remove.

This specification should be interpreted as a robustness diagnostic rather than as a structural model of welfare reform. The interaction of year effects with $W_i$ asks whether the EITC coefficient is robust to allowing aggregate shocks in the 1990s to vary with pre-reform reliance on welfare. Because $W_i$ is estimated using only pre-reform data, it is predetermined with respect to the post-reform employment changes analyzed here.[5] Interacting year effects with $W_i$ allows any aggregate shock of the 1990s—the administrative changes of welfare reform, but equally macroeconomic conditions, minimum-wage increases, or Medicaid expansions—to affect employment in proportion to pre-reform welfare reliance. The interaction is deliberately agnostic about which shock matters; it asks only whether the EITC estimates are identified by EITC variation rather than by the correlation of EITC treatment with baseline welfare exposure. An effect identified by EITC variation would survive the adjustment. Identification of the EITC's effect would require EITC variation orthogonal to pre-reform welfare exposure; the simulated-EITC exercise in Appendix E.6 isolates exactly that residual variation and finds it explains none of the placebo estimates.

At the same time, one might worry that this approach effectively "over-controls" by absorbing part of a genuine EITC effect if the policy's impact happened to be correlated with pre-reform welfare exposure. The next section addresses this concern.

[5]The estimation window includes 1993, by which time a few states had implemented AFDC waivers. Any such contamination of the base year would attenuate the measured exposure gradient, working against the findings here; Appendix D shows the gradient is national rather than waiver-driven.

# 4 Differential Exposure to Welfare Reform as a Source of Bias

The previous section raised the concern that the welfare-exposure control might absorb part of a true EITC effect rather than remove a confound. Two exercises address it.

First, I examine employment trajectories within subgroups defined by their predicted welfare exposure. Mothers with similar pre-reform exposure follow similar paths regardless of EITC treatment status, implying that much of the difference across treatment and control groups reflects composition rather than EITC treatment. Second, I conduct falsification tests—placebo comparisons—that assign artificial "treatment" and "control" labels to subgroups who in reality faced the same EITC schedule. These tests show that the canonical DiD estimator generates spurious effects that scale precisely with differences in pre-reform welfare exposure. Together, this evidence demonstrates that the conventional DiD estimates are not identified from genuine treatment–control contrasts, but instead reflect compositional imbalance in exposure to welfare reform. Appendix C provides a formal decomposition of the DiD estimator, showing explicitly how differences in subgroup composition mechanically generate the observed estimates.

## 4.1 Evidence of Non-Parallel Trends

To assess whether the observed treatment effects reflect the EITC or differences in welfare exposure, I begin by comparing employment trends among women with similar predicted pre-reform welfare exposure. If the EITC were driving the results, then differences between treatment and control groups should persist even after conditioning on welfare exposure. By contrast, if exposure to welfare reform explains the results, then women with similar pre-reform propensities should follow parallel paths regardless of EITC eligibility.

The top panel of Figure 3 plots employment trends of single mothers, disaggregated by predicted pre-reform welfare exposure (Lowest 60%, 60–80%, Top 20%), separately for mothers of one child (solid) and two or more children (dashed). The figure shows, first, that groups with higher pre-reform welfare exposure experienced much larger employment increases after 1993, while groups with low pre-reform exposure saw little change. Second, conditional on pre-reform exposure, mothers with one versus two or more children follow nearly identical trajectories in the lower-exposure bins, with a residual gap in the high-exposure bin, where the one-child cell is small and noisy (Appendix E). Because EITC incentives and welfare exposure remain correlated within bins, this comparison is corroborative rather than dispositive; the identification weight rests on the falsification tests of Section 4.2, which compare groups facing the same statutory EITC schedule.

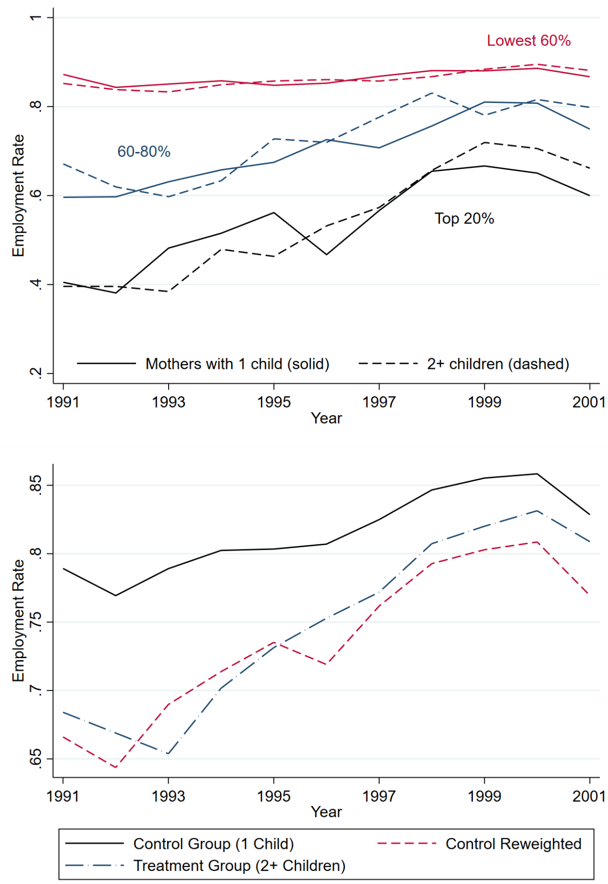
The second panel plots the employment rates of mothers of one child (the solid black line) and mothers of 2+ children (the blue long-dashed line), together with mothers of one child reweighted to match the predicted pre-reform welfare exposure of the treatment group. Once reweighted, the one-child and two-or-more-child lines follow nearly the same path, and the large relative employment increase among mothers of 2+ children disappears.

The composition of the two groups thus explains why conventional DiD estimates are large. Mothers with exactly one child are disproportionately drawn from the lowest-exposure strata (77% in the lowest 60%, 16% in the 60–80%, and only 7% in the top 20%). By contrast, mothers with two or more children are much more concentrated in high-exposure strata (47% in the lowest 60%, 23% in the 60–80%, and 30% in the top 20%). Because employment surged only among high-exposure groups, and those groups are overrepresented in the treatment group, the difference-in-differences estimator mechanically attributes those gains to the EITC. Were the two groups balanced on pre-reform welfare exposure, the treatment effect of the EITC would essentially disappear, as Figure 3 shows.[6]

Figure 3 is suggestive but not decisive because balancing on welfare exposure does not fully separate welfare exposure from EITC treatment. A defender of the EITC interpretation could argue that one-child mothers responded more strongly per dollar of EITC generosity—

[6] Although mothers with two or more children are substantially more likely to be in high-exposure cells, the family-size groups have meaningful overlap in predicted welfare exposure; see Appendix Figure A3.

Figure 3: Employment Rates of Single Mothers by Predicted Pre-Reform Welfare Use, 1991–2001




Notes: Top panel: Solid lines show mothers with one child; dashed lines show mothers with two or more children. Groups are stratified by quintiles of predicted pre-reform welfare exposure (bottom three quintiles combined as "Lowest 60%"). Conditional on pre-reform welfare exposure, employment trajectories are similar across treatment and control groups, diverging mainly in the highest-exposure cells. Bottom panel: The black line shows raw employment of mothers with one child (control); the red dashed line reweights this group to match the welfare-exposure distribution of mothers with two or more children; the blue dashed line shows raw employment of mothers with two or more children (treatment). After reweighting, the employment paths of treatment and control groups are nearly identical. Sample: Single mothers ages 20–50, March CPS 1991–2001.

especially in high-exposure bins—so that their employment kept pace with mothers with two or more children despite receiving a smaller statutory expansion. Appendix E shows that this is not a mechanical consequence of eliminating EITC variation: the EITC incentive contrast remains positive within every welfare-exposure bin. More importantly, the falsification tests in Section 4.2 avoid this concern altogether by comparing groups facing the same statutory EITC schedule. In those comparisons, residual simulated EITC incentive variation has no explanatory power once welfare exposure is accounted for (Appendix Table A9).

## 4.2 Falsification Tests of the Canonical Design

A complementary way to show that the canonical DiD estimator is biased is to run placebo tests that ask whether the estimator produces "effects" in contexts where, by construction, the true treatment effect is zero. If the estimator systematically generates spurious effects related to pre-reform welfare exposure, this shows that the identifying assumptions are violated and the failure to control for welfare exposure is the source of the bias. To see this formally, consider a simplified model

$$y_{it} = \beta\, D_{it} + X_{it}\delta + \kappa\,(\text{Post}_t \times Z_i) + \epsilon_{it},$$

where $D_{it} = \text{Post}_t \times T_i$ is the difference-in-differences treatment, $T_i$ indicates the treatment group, and $Z_i$ is a time-invariant factor correlated with treatment. A level term in $Z_i$ with a time-constant effect would difference out and cannot generate difference-in-differences bias; here the bias operates through the interaction with the post-reform period. Taking pre-to-post first differences sweeps out the time-invariant terms, so that, partialling out the controls, the change in employment is

$$\Delta y_i = \beta\, T_i + \kappa\, Z_i + \Delta\epsilon_i.$$

If $Z_i$ is omitted, the estimated treatment effect reflects omitted-variables bias:

$$\hat{\beta} = \beta + \kappa \frac{\operatorname{cov}(T_i, Z_i)}{\operatorname{var}(T_i)},$$

which, because $T_i$ is a binary group indicator, reduces to

$$\hat{\beta} = \beta + \kappa \Delta Z, \tag{2}$$

the effect of $Z$ on the post-period employment change times the mean difference in $Z$ across groups. In the EITC context, the hypothesized omitted factor is welfare exposure: $Z_i = W_i$ and $\Delta Z = \Delta W$, the difference in mean pre-reform welfare exposure between the compared groups.

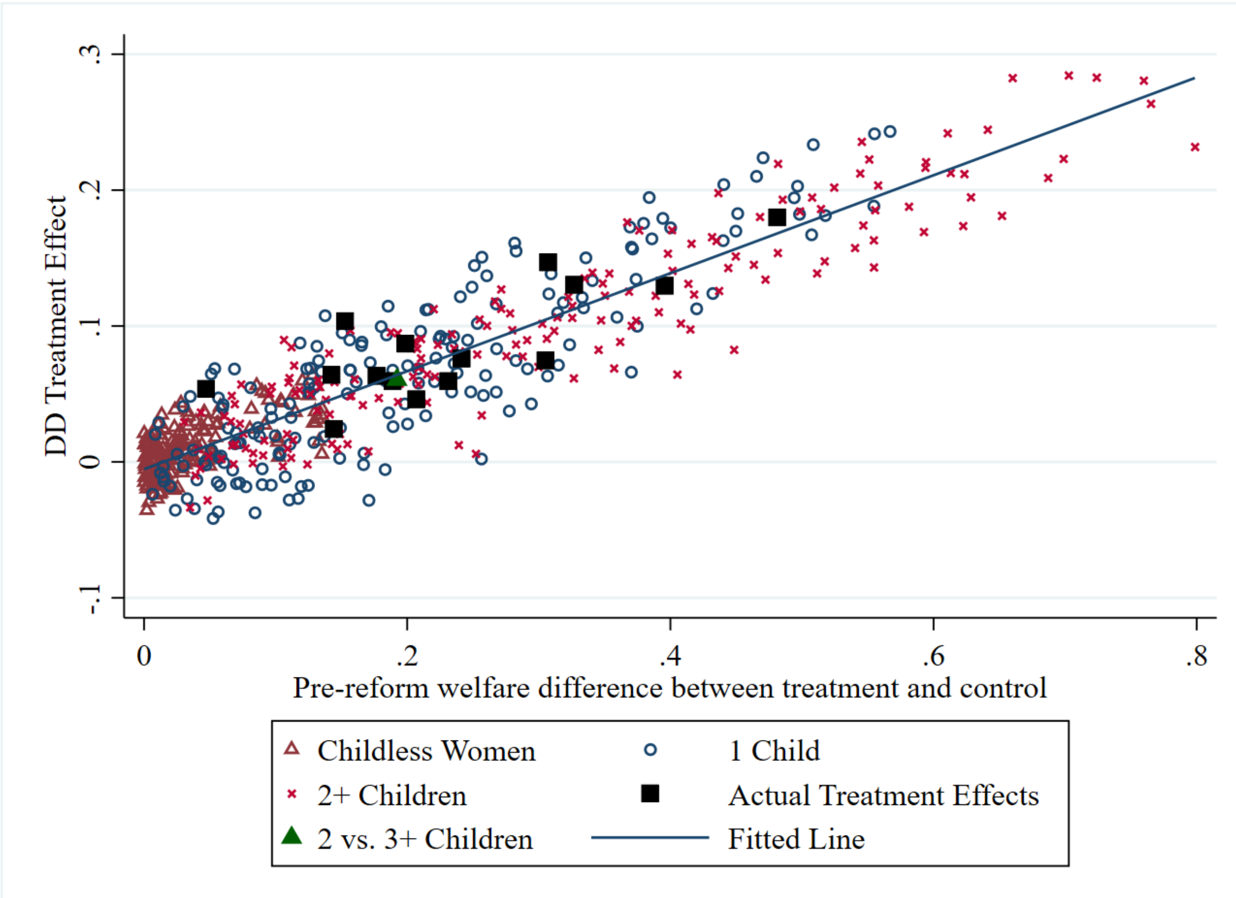
This framing clarifies the logic of the design and gives these comparisons the structure of a falsification test. Under the canonical design's identifying assumption, groups facing the same EITC schedule should exhibit no relative employment change after conditioning on the included covariates: the contrast holds the statutory schedule fixed. Cells within a schedule can still differ in dollar incentives because their earnings distributions differ; that residual channel is tested directly below (Appendix Table A9) and explains none of the estimates. A nonzero placebo estimate is therefore not a "heterogeneous effect"—it is a direct measure of bias, and Equation (2) shows that its magnitude identifies the omitted confounder. The results below establish four facts with this structure: the placebo estimates are large and linear in the predetermined welfare-exposure gap; they are equally present for childless women; they are reproduced by an identification strategy built on an entirely different treatment margin (welfare time limits, Appendix G); and the same variable that predicts them eliminates them when included as a control (Appendix Table A8). Unlike the classic case of unobserved ability in wage regressions [e.g., Angrist and Pischke, 2009], here the relevant confounder—exposure to welfare reform—is measurable, but deliberately excluded under the assumption that other controls (e.g., state-by-year fixed effects) are sufficient.

**Design and Implementation.** I form treatment and control groups that are equally exposed to the EITC, so any estimated difference-in-differences must be spurious. To do this, I restrict attention to mothers with the same EITC treatment status (e.g., mothers with one child, mothers with two or more children, or childless women). Within each group, I divide individuals into subgroups based on pre-reform characteristics. In particular, using 1991–1993 data, I estimate predicted probabilities of (i) welfare receipt, (ii) employment, and (iii) welfare receipt without employment from probit models, and (iv) predicted earnings from OLS regressions, with covariates for education, race, marital status, age of youngest child, state of residence, and a quartic in age. I then sort individuals into deciles of each predicted variable within each EITC treatment group.

For each EITC group, I assign one decile as a "placebo treatment" and another as a "placebo control," and re-estimate the DiD regression. Because these subgroups face the same EITC policy schedule, the true treatment effect should be zero. Repeating this procedure across all possible decile pairings yields 90 placebo estimates per propensity measure within each EITC group; the exhibits restrict attention to pairs with a nonnegative welfare-exposure gap—each pairing otherwise enters twice with opposite signs—leaving 540 placebo estimates. For each regression, I also record the difference in predicted welfare exposure between the two placebo groups as a measure of imbalance in the suspected confounder.

I compare these placebo estimates to the actual DiD estimates from Schanzenbach and Strain [2021], Meyer and Rosenbaum [2001], Hoynes and Patel [2018], and Michelmore and Pilkauskas [2021]. Because each paper compares multiple groups (e.g., mothers of 2+ children, mothers of one child, and childless women), I form and estimate all the pairwise comparisons that comprise each estimate following Goodman-Bacon [2021]. These can then be directly compared to the placebo estimates.

Figure 4: Placebo and Actual DiD Estimates by Pre-Reform Welfare Exposure




Notes: Each hollow symbol represents a placebo DiD estimate comparing two decile groups within the same EITC treatment category: triangles for childless women, circles for mothers with one child, and "×" for mothers with two or more children. Deciles are formed within each category by predicted pre-reform welfare receipt, employment, welfare receipt without employment, and predicted earnings. By construction, all placebo comparisons have a true treatment effect of zero. The horizontal axis shows the difference in pre-reform welfare exposure between the placebo "treatment" and "control" groups. Black squares are actual DiD estimates from pairwise comparisons in Schanzenbach and Strain [2021], Hoynes and Patel [2018], Meyer and Rosenbaum [2001], and Michelmore and Pilkauskas [2021]. The solid green triangle compares mothers of three or more children to mothers of exactly two, who faced an identical EITC schedule after 1993. The fitted line is estimated on placebo estimates only (slope = 0.36, s.e. = 0.01, with a small negative intercept). Sample: Single women ages 20–50, March CPS 1991–2001.

**Results.** Figure 4 shows two clear findings of this exercise. First, placebo estimates are often large and economically meaningful: the DiD estimator finds "effects" where none exist, including among childless women, who faced essentially no EITC—the placebo slope estimated on childless-women pairs alone is 0.35 (0.04), indistinguishable from the pooled slope of 0.36. Second, the magnitude of these spurious effects is linear in the pre-reform welfare-use gap between groups. When placebo groups are similar, estimates are near zero; when imbalanced, estimates are large. The actual DiD estimates from the canonical literature fall exactly on this same fitted line. So does a comparison the EITC schedule itself renders nearly treatment-free: mothers of three or more children versus mothers of exactly two, who faced an identical credit schedule after 1993, produce an estimated "effect" of 6.1 percentage points—close to the 6.4 points the fitted placebo line implies at their pre-reform welfare-exposure gap of 0.192. Far from being reassuring about the design, the two-versus-three-plus contrast exhibits the same proportional bias, and the fanning out of estimated effects by family size emerges mechanically in the simulation of Appendix F without any EITC response.

In other words, knowing only the placebo results from any single treatment group and the baseline welfare-use gap is enough to predict the published results—without needing any data on post-reform outcomes from other treatment groups. Taken together, these findings imply that the canonical DiD estimates are not identified from genuine treatment–control contrasts in EITC exposure. They are mechanically generated by differences in baseline exposure to welfare reform. Appendix Table A8 reinforces this interpretation: the slope of the placebo estimate with respect to the welfare-exposure gap is large in the baseline specification but falls close to zero once Year $\times$ predicted pre-reform welfare exposure is added as a control. The variable that predicts the bias eliminates it once included.

A remaining objection is that within-group placebo comparisons might capture genuine EITC responses if EITC incentives vary across cells within a family-size group—for example, if low-earning cells gained more from the expansion. Appendix Table A9 addresses this interpretation directly. For every placebo pair, I compute, alongside the welfare-exposure gap $\Delta W$, the corresponding contrast in a simulated EITC that evaluates each law year's credit at fixed earnings of each group (Appendix E.6). Regressed alone, the incentive contrast appears to predict the placebo estimates, but only because it is correlated with welfare exposure; conditional on $\Delta W$, its coefficient is economically negligible and slightly negative, it adds nothing to explained variation ($R^2$ rises from 0.850 to 0.852), and the coefficient on $\Delta W$ is unchanged. The placebo estimates cannot be reinterpreted as responses to statutory incentive variation across cells.

## 4.3 Corroboration: Welfare Time Limits

A useful corroboration comes from a related literature on welfare time limits. Grogger [2004] estimates the effect of newly-imposed five-year time limits on welfare use, exploiting variation in the age of the youngest child: families with younger children face a binding constraint (they could exhaust benefits before the child ages out), while families whose youngest child is a teenager do not. Crucially, this identification strategy relies on entirely different treatment variation than the EITC studies—age of youngest child rather than number of children—and the theoretical mechanism (precautionary savings of welfare benefits) is unrelated to EITC incentives. Yet the estimator is equally exposed to the same potential confound, because mothers of younger children also had higher pre-reform welfare exposure than mothers of teenagers.

Appendix G shows that the time-limit estimates exhibit the same pattern documented above. The DiD estimates fall on the same fitted line as both the EITC estimates and the placebo estimates (Figure A8), and controlling for Year $\times$ pre-reform welfare exposure attenuates them toward zero (Figure A9). The fact that an entirely separate identification strategy, based on different treatment groups and a different policy mechanism, produces the same spurious pattern corroborates the conclusion that the bias arises from differential exposure to welfare reform rather than from any particular policy's true effect.

# 5 Preventing Bias from Omitted Variables with Machine Learning

One way to prevent this type of error in future research is to search systematically for potential confounders before the treatment coefficient is estimated and interpreted, supplementing researcher judgment about covariate selection.

A pragmatic approach is to use machine learning (ML) to search for omitted variables that may bias the estimator. Conventional approaches to covariate selection are often ad hoc, relying on researcher discretion about which variables to include or how to specify them. Even more systematic approaches, such as matching or synthetic control methods, typically rely only on pre-reform data and exclude information from the post-reform period. As a result, they may miss variables that play little role in pre-trends but become important confounders in the post-reform period (like exposure to welfare reform). A more systematic alternative is to let the data themselves highlight which variables predict outcomes across groups.

I demonstrate this approach using the simple lasso algorithm. The lasso selects a sparse set of predictors from a large pool of covariates. In this case, I form a large set of potential baseline covariates interacted with year effects (such as demographics, predicted pre-reform welfare exposure, employment, and wages). Concretely, I train the lasso first on one group (say, mothers with one child) to predict employment outcomes over time, and then generate out-of-sample predictions for the other group (say, mothers with two or more children).

Training within a single group withholds any information on treatment status. The cross-group predictions therefore represent the algorithm's best guess at the target group's path, constructed without any data from that group—a counterfactual net of the training group's own (smaller) treatment.

If the EITC truly caused the divergence between groups, these cross-predictions should systematically miss—predicted employment too high for controls and too low for the more-treated group. If instead the predictions track observed outcomes closely, that suggests the divergence is fully explained by observable confounders—such as pre-reform welfare exposure—that the lasso has identified as predictive.

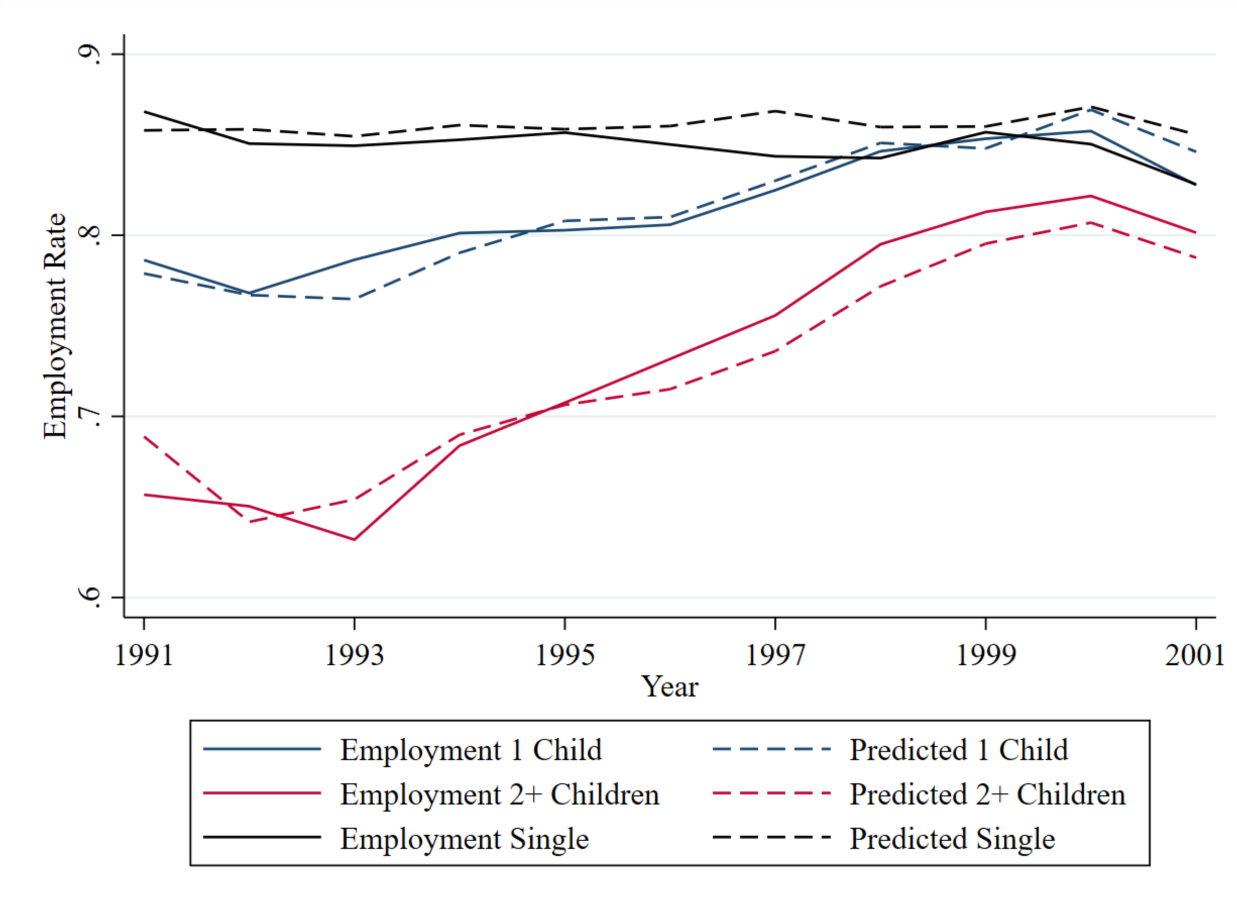
This approach maps directly onto the familiar graphical assessment of the parallel trends assumption: plotting actual and predicted outcomes side by side shows whether treatment and control groups would have followed similar paths absent treatment. Figure 5 illustrates this, comparing actual employment trends (solid lines) with the out-of-sample ML predictions (dashed lines). Because predictions for each group are generated exclusively from data on a different group, they embed no information about the target group's own treatment.

First, Figure 5 shows that the out-of-sample predictions closely track the actual outcomes. This is evidence against a large EITC treatment effect: the counterfactual constructed from the other group matches realized outcomes closely. If the EITC were the primary driver, one would expect large, systematic forecast errors—predicted employment that is too high for controls and too low for the more-treated group. Instead, the alignment between actual and predicted outcomes supports the conclusions from earlier sections and provides a direct graphical check of the parallel trends assumption.

Second, the ML procedure consistently selects interactions of time with pre-reform welfare participation (or correlates of welfare participation, like age of the youngest child) as important predictors within groups. These are the same confounders tied to exposure to welfare reform identified in Sections 3 and 4. With these covariates in hand, one could have generated the correct specification ex ante or, at minimum, flagged that the parallel-trends assumption was unlikely to hold post-reform.

Finally, the cross-predictions serve as a transparent design diagnostic. Even when pre-trends appear parallel, the post-reform out-of-sample predictions make clear that treatment and control groups were unlikely to remain parallel absent treatment—both because higher-welfare-use subgroups exhibit distinct within-group trends and because employment among childless women was already near its ceiling.

Figure 5: Actual and Predicted Out-of-Sample ML Employment Rates




Notes: Solid lines show actual annual employment rates; dashed lines show out-of-sample predictions from a lasso model with cross-validation. Covariates include demographics, age polynomials, and interactions of year with predicted pre-reform welfare exposure, employment, and wages. Predictions for mothers of two or more children and childless women are estimated exclusively from data on mothers of one child; predictions for mothers of one child are estimated from data on mothers of two or more children. Because predictions are trained on a different group, they embed no information about the target group's treatment and represent the algorithm's best guess of its counterfactual path. The close alignment between actual and predicted outcomes leaves little room for an EITC treatment effect. Sample: Single women ages 20–50, March CPS 1991–2001.

# 6 Conclusion

This paper reexamines the canonical difference-in-differences evidence that the 1993 EITC expansion increased employment among single mothers. The central finding is that these estimates are biased by differential exposure to welfare reform. Treatment and control groups defined by the number of children differ systematically in predicted pre-reform welfare exposure. In the replicated specifications, allowing employment trends to vary with that exposure eliminates the robust positive EITC effects. Reweighting the groups to balance welfare exposure removes the treatment-control divergence, while placebo comparisons among groups facing the same statutory EITC schedule reproduce the canonical estimates. The same exposure gradient appears in estimates of welfare time limits, and a machine-learning approach selects the same exposure interactions and would have flagged the failure of parallel trends. Taken together, the evidence points to welfare reform as the source of the employment divergence that the canonical designs attribute to the EITC.

These findings are specific to estimates of the employment effects of the 1993 expansion; whether the same identification problem affects other outcomes is unclear. Because the 1993 expansion occupies a central place in the literature, however, the results raise the possibility that studies using the same identifying variation to estimate effects on poverty [Hoynes and Patel, 2018], child achievement [Dahl and Lochner, 2012], maternal health [Evans and Garthwaite, 2014], infant health [Hoynes et al., 2015], and other outcomes may also capture differential exposure to welfare reform rather than the EITC alone.

These findings bear on current policy debates, in which the 1990s are routinely cited as evidence that work-contingent tax credits drew single mothers into employment. The evidence in this paper suggests that the 1990s comparisons provide little support for employment effects of work-contingent credits. Recent evidence and projections reach varying conclusions: Goldin et al. [2024] find little maternal labor-supply response to extending California's child tax credit to nonworkers, while Corinth et al. [2021] project employment losses from a fully extended credit. This disagreement helps explain why the 1990s evidence has been asked to carry so much weight. Debates over work requirements and refundable child benefits should turn on research designs that vary those features directly. The 1990s are better read as a story about welfare reform than about the Earned Income Tax Credit.

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# A Summary Statistics

The analysis uses data from the March Current Population Survey (CPS) Annual Social and Economic Supplement, obtained via IPUMS [Flood et al., 2024], for the years 1991–2001. The sample is restricted to unmarried women ages 20–50. Mothers whose youngest child is older than 18 are dropped, as are observations with implausible age gaps between the mother and her youngest child; childless women are retained as a comparison group. Income variables are deflated to constant dollars using the CPI.

The primary outcome is an indicator for any positive earnings in the previous year. Covariates include a quartic polynomial in age, indicators for education (less than high school, high school, some college, college or more), race, marital status, age of youngest child, and state of residence. All regression specifications include state-by-year fixed effects.

EITC treatment variables are constructed following each of the four replicated studies and merged from external sources: the Schanzenbach and Strain [2021] post-1993 treatment indicator (with phase-in adjustments), the Hoynes and Patel [2018] simulated EITC instrument (incorporating state EITC programs), the Meyer and Rosenbaum [2001] average taxes-if-work variable by number of children, and the Michelmore and Pilkauskas [2021] average EITC by state, year, and number of children. State welfare time-limit indicators are merged separately for the analysis in Appendix G.

Table A1 reports summary statistics for the pooled sample. Table A2 shows annual means of key outcomes and treatment variables by number of children.

Table A1: Summary Statistics

| Variable | Mean | SD |
|---|---|---|
| Earned any income | 0.82 | 0.38 |
| Welfare participation | 0.10 | 0.29 |
| Earnings ($) | 20,898 | 25,630 |
| Age | 31.97 | 8.86 |
| No children | 0.62 | 0.49 |
| One child | 0.17 | 0.37 |
| Two or more children | 0.21 | 0.41 |
| High school | 0.30 | 0.46 |
| Some college | 0.35 | 0.48 |
| College graduate | 0.22 | 0.41 |
| Observations | 150,433 | |

Table A2: Panel Outcomes and Policy Variables by Year

| | 1991 | 1992 | 1993 | 1994 | 1995 | 1996 | 1997 | 1998 | 1999 | 2000 | 2001 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| *Panel A: Childless Women* | | | | | | | | | | | |
| Work | 0.87 | 0.85 | 0.85 | 0.85 | 0.86 | 0.85 | 0.84 | 0.84 | 0.86 | 0.85 | 0.83 |
| Welfare | 0.03 | 0.04 | 0.03 | 0.02 | 0.02 | 0.02 | 0.02 | 0.01 | 0.01 | 0.01 | 0.01 |
| SWS Treatment | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| MR Tax if Work | 2,973 | 2,935 | 2,934 | 2,901 | 2,899 | 2,907 | | | | | |
| HP Sim EITC | 0 | 0 | 0 | 40 | 40 | 40 | 40 | 41 | 40 | 39 | 40 |
| *Panel B: Mothers with One Child* | | | | | | | | | | | |
| Work | 0.79 | 0.77 | 0.79 | 0.80 | 0.80 | 0.81 | 0.82 | 0.85 | 0.85 | 0.86 | 0.83 |
| Welfare | 0.23 | 0.24 | 0.24 | 0.22 | 0.19 | 0.18 | 0.15 | 0.12 | 0.10 | 0.08 | 0.07 |
| SWS Treatment | 0 | 0 | 0 | 0.92 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| MR Tax if Work | 1083 | 1001 | 955 | 702 | 633 | 609 | | | | | |
| HP Sim EITC | 625 | 693 | 729 | 957 | 1010 | 1013 | 924 | 948 | 941 | 917 | 927 |
| MP Average EITC | 711 | 781 | 825 | 1092 | 1155 | 1160 | 1172 | 1204 | 1193 | 1166 | 1185 |
| *Panel C: Mothers with Two or More Children* | | | | | | | | | | | |
| Work | 0.66 | 0.65 | 0.63 | 0.68 | 0.71 | 0.73 | 0.76 | 0.80 | 0.81 | 0.82 | 0.80 |
| Welfare | 0.38 | 0.37 | 0.39 | 0.35 | 0.31 | 0.29 | 0.25 | 0.20 | 0.16 | 0.14 | 0.11 |
| SWS Treatment | 0 | 0 | 0 | 0.50 | 0.78 | 1 | 1 | 1 | 1 | 1 | 1 |
| MR Tax if Work | 651 | 554 | 501 | 19 | -291 | -548 | | | | | |
| HP Sim EITC | 633 | 707 | 745 | 1194 | 1457 | 1670 | 1440 | 1473 | 1469 | 1436 | 1452 |
| MP Average EITC | 780 | 865 | 916 | 1453 | 1769 | 2014 | 2028 | 2079 | 2060 | 2024 | 2048 |

Notes: Weighted annual means (March CPS supplement weights). “Work” is the share with any earnings during the year; “Welfare” is the share receiving cash welfare. “SWS Treatment” is the phase-in-adjusted post-1993 treatment indicator of Schanzenbach and Strain [2021]. “MR Tax if Work” is the Meyer and Rosenbaum [2001] average net tax liability if working, in dollars; negative values indicate net transfers. The variable is defined through 1996, the end of their sample period. “HP Sim EITC” and “MP Average EITC” are the Hoynes and Patel [2018] simulated EITC and the Michelmore and Pilkauskas [2021] average EITC, in dollars. Sample: single women ages 20–50, March CPS 1991–2001.

# B Replication and Comparison to the Literature

## B.1 Data and Sample

The sample and variable definitions are identical to those in Appendix A: unmarried women ages 20–50 from the March CPS, 1991–2001, with employment (positive earnings last year) as the dependent variable, standard demographic covariates, and state-by-year fixed effects in all specifications.

## B.2 Treatment Variables

I focus on four influential papers that are representative of the published evidence on the EITC's labor supply effects, use common and well-known data from the March CPS, but differ in their parameterization of the EITC, the subpopulations studied, and whether they incorporate only federal or both federal and state EITC expansions.

**Schanzenbach and Strain [2021] (SWS).** The EITC expansion is represented as a dummy variable equal to zero before 1994; equal to 0.92 for mothers of one child and 0.5 for mothers of two or more children in 1994 (reflecting partial phase-in); equal to 1 for mothers of one child and 0.78 for mothers of two or more children in 1995; and equal to 1 for both groups thereafter. Childless women serve as the control group. Policy variation is at the national level, and the authors report estimates for subpopulations by educational attainment (less than high school, high school only).

**Meyer and Rosenbaum [2001] (MR).** The EITC's effect is captured through the variable "taxes if work," which measures the average level of taxes (net of the EITC) if working, varying by year and number of children (0, 1, or 2+). This specification exploits variation at the year $\times$ number-of-children level and was estimated over the period 1984–1996, specifically excluding post-1996 welfare policy changes.

**Hoynes and Patel [2018] (HP).** The EITC is represented as a simulated dollar value in each state (incorporating any state-level EITC) by year and by number of children (0 to 3). This approach captures both federal expansions and cross-state variation from state EITC programs. The authors report estimates for the full sample and by educational attainment.

**Michelmore and Pilkauskas [2021] (MP).** This study examines heterogeneity in the EITC's effect by the age of the mother's youngest child. The EITC is modeled as a simulated value at the state $\times$ year $\times$ number-of-children level (for mothers with 1–3 children; childless women are excluded), and this treatment variable is interacted with dummies for age of youngest child (0–2, 3–5, 6–12, 13+). The analysis spans 1990–2016.

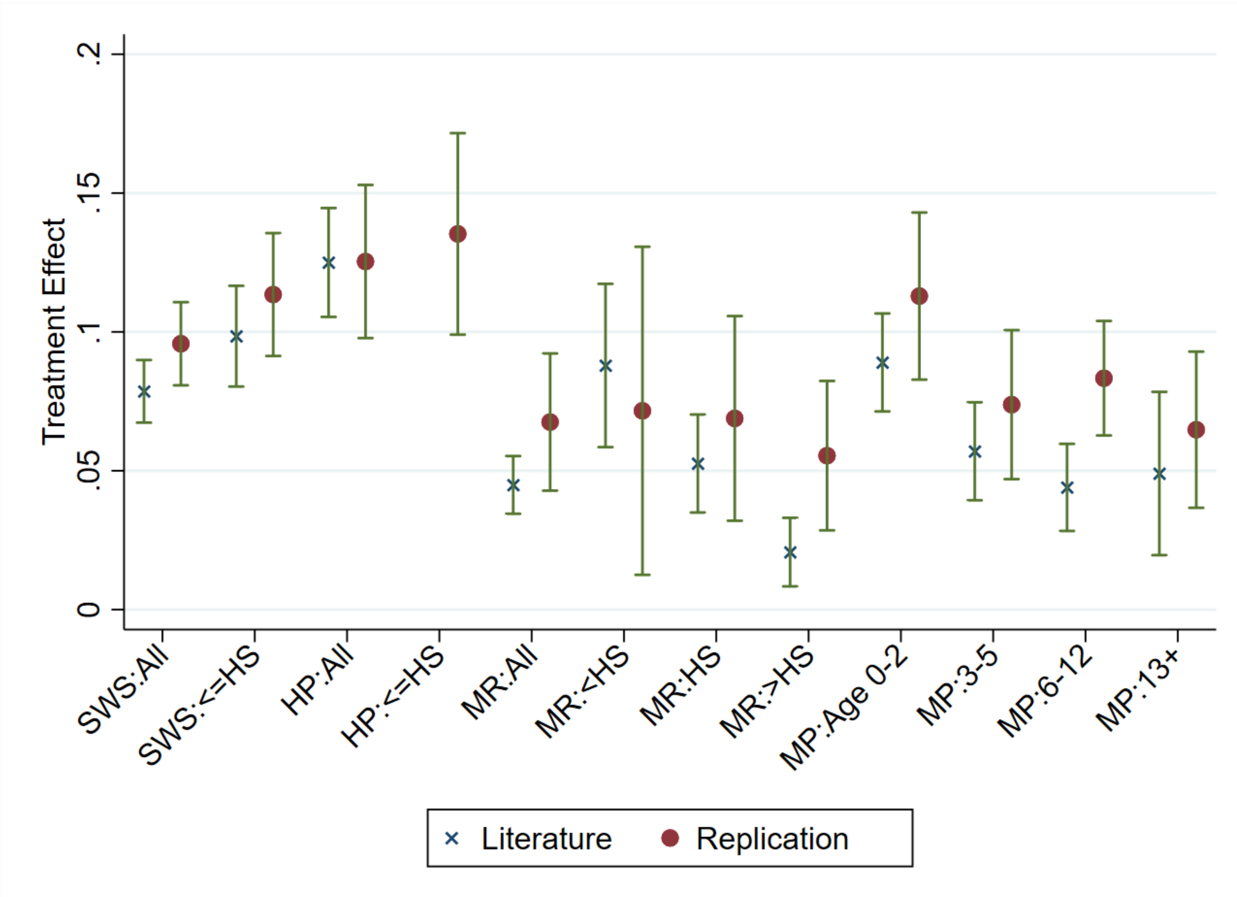
## B.3 Baseline Replication

Figure A1 compares my re-estimated EITC effects to the published coefficients across these four studies. Red dots are my replication; blue crosses are the published literature; vertical bars are 95% CIs for the replication. Despite differences in sample periods, treatment parameterization, and subsamples, the replication closely matches the published magnitudes and statistical significance patterns.

With the exception of Meyer and Rosenbaum [2001], each specification is estimated over the period from 1991–2001. The replication of Meyer and Rosenbaum [2001]'s specification is 1991–1996.

Whereas Michelmore and Pilkauskas [2021] use "worked last week" as the outcome, my replication uses "earned any income last year"—yet the scales remain comparable. Overall, the replication corroborates the canonical estimates and thus provides a credible baseline for the controlled specifications that add $\text{Year}_t \times W_i$. Table A3 reports the underlying coefficient estimates and standard errors.

Figure A1: Published vs. Replicated EITC Treatment Effects




Notes: Blue "×" markers are point estimates reported in the published literature; red dots are my replication with 95% confidence intervals (vertical bars). SWS = Schanzenbach and Strain [2021]; HP = Hoynes and Patel [2018]; MR = Meyer and Rosenbaum [2001]; MP = Michelmore and Pilkauskas [2021]. Specifications differ in EITC parameterization (dummy, simulated value, taxes-if-work), sample restrictions (by education), and whether state EITC programs are incorporated. All replications use March CPS 1991–2001, except MR which uses 1991–1996. Standard errors clustered at the state-by-year level.

Table A3: Replication of Published EITC Estimates on Employment

| | Literature | Replication |
|---|---|---|
| *Panel A: Schanzenbach and Strain* | | |
| All women | 0.0786 | 0.0957 |
| | ( 0.0058) | ( 0.0076) |
| ≤ High school | 0.0985 | 0.1134 |
| | ( 0.0093) | ( 0.0113) |
| *Panel B: Hoynes & Patel* | | |
| All women | 0.1250 | 0.1253 |
| | ( 0.0100) | ( 0.0141) |
| ≤ High school | | 0.1353 |
| | | ( 0.0185) |
| *Panel C: Meyer & Rosenbaum* | | |
| All women | 0.0449 | 0.0675 |
| | ( 0.0053) | ( 0.0126) |
| < High school | 0.0879 | 0.0716 |
| | ( 0.0150) | ( 0.0301) |
| High school | 0.0526 | 0.0688 |
| | ( 0.0090) | ( 0.0188) |
| > High school | 0.0207 | 0.0554 |
| | ( 0.0063) | ( 0.0137) |
| *Panel D: Michelmore & Pilkauskas* | | |
| Age 0–2 | 0.0890 | 0.1129 |
| | ( 0.0090) | ( 0.0154) |
| Age 3–5 | 0.0570 | 0.0738 |
| | ( 0.0090) | ( 0.0137) |
| Age 6–12 | 0.0440 | 0.0833 |
| | ( 0.0080) | ( 0.0105) |
| Age 13+ | 0.0490 | 0.0648 |
| | ( 0.0150) | ( 0.0143) |

Notes: Each row reports a point estimate and standard error (in parentheses) from the published literature and my replication. Panel A: post-1993 EITC treatment dummy (2+ children × post). Panel B: simulated EITC instrument ($1,000s). Hoynes and Patel report employment estimates (in an online appendix) for a some-college-or-less sample; no published coefficient is directly comparable to the high-school-or-less subsample, so only the replication is shown. Panel C: taxes-if-work variable ($1,000s); sample restricted to years before 1997. Panel D: average EITC (by state, year, and number of children) interacted with age-of-youngest-child group dummies. All regressions include demographic controls (age quartic, education, race, marital status, number of children, age of youngest child) and state-by-year fixed effects, weighted using CPS survey weights with standard errors clustered at the state × year level.

## B.4 Controlling for Welfare Exposure

Table A4 reports the results of adding Year $\times$ predicted pre-reform welfare exposure interactions to each specification. Across all studies and subsamples, the EITC coefficients attenuate toward zero, while the welfare $\times$ year interaction is large, positive, and highly significant—indicating that the original estimates load on differential welfare-reform exposure rather than on EITC variation.

Table A4: EITC Estimates Controlling for Pre-Reform Welfare Exposure

| | (1) EITC Coefficient | (2) Welfare × Year Coefficient | (3) Mean Welfare Exposure | (4) Implied Δ Employment | (5) Actual Δ Employment |
|---|---|---|---|---|---|
| *Panel A: Schanzenbach and Strain* | | | | | |
| All women | 0.0126 | 0.5373 | 0.086 | 0.046 | 0.055 |
| | ( 0.0063) | ( 0.0552) | | | |
| ≤ High school | 0.0097 | 0.5291 | 0.157 | 0.083 | 0.094 |
| | ( 0.0112) | ( 0.0632) | | | |
| *Panel B: Hoynes & Patel* | | | | | |
| All women | 0.0134 | 0.5463 | 0.086 | 0.047 | 0.055 |
| | ( 0.0105) | ( 0.0543) | | | |
| ≤ High school | -0.0137 | 0.5625 | 0.157 | 0.088 | 0.094 |
| | ( 0.0170) | ( 0.0632) | | | |
| *Panel C: Meyer & Rosenbaum* | | | | | |
| All women | 0.0103 | 0.2329 | 0.086 | 0.020 | 0.024 |
| | ( 0.0127) | ( 0.0542) | | | |
| < High school | 0.0135 | 0.2152 | 0.280 | 0.060 | 0.025 |
| | ( 0.0435) | ( 0.1335) | | | |
| High school | -0.0487 | 0.5860 | 0.104 | 0.061 | 0.032 |
| | ( 0.0246) | ( 0.1314) | | | |
| > High school | 0.0127 | 0.4969 | 0.031 | 0.015 | 0.009 |
| | ( 0.0152) | ( 0.1191) | | | |
| *Panel D: Michelmore & Pilkauskas* | | | | | |
| Age 0–2 | 0.0102 | 0.5595 | 0.340 | 0.190 | 0.230 |
| | ( 0.0167) | ( 0.0638) | | | |
| Age 3–5 | 0.0023 | | 0.222 | 0.124 | 0.124 |
| | ( 0.0142) | | | | |
| Age 6–12 | 0.0211 | | 0.154 | 0.086 | 0.112 |
| | ( 0.0112) | | | | |
| Age 13+ | 0.0121 | | 0.087 | 0.049 | 0.057 |
| | ( 0.0137) | | | | |

Notes: Columns (1)–(2) report coefficient estimates and standard errors (in parentheses) from regressions that add Year × predicted pre-reform welfare exposure interactions to the baseline specifications in Table A3. Column (1): EITC coefficient after controlling for welfare exposure. Column (2): coefficient on the interaction of predicted pre-reform welfare use with the post-period year indicator (1999 for Panels A, B, and D; 1996 for Panel C); regressions include the full set of year interactions but only the key post-period interaction is shown. Column (3): weighted mean of predicted welfare exposure (Pr(on welfare and not working | X)) in each specification's sample. Column (4) = (2) × (3): the implied average employment increase attributable to welfare reform for the sample. Column (5): actual change in the employment rate from 1993 to the post period (1999 or 1996). In Panel D the specification includes a single Year × welfare-exposure interaction common to all age groups; its coefficient appears once, in the Age 0–2 row, and Column (4) applies it to each age group's mean exposure. See Table A3 notes for other specification details.

# C Decomposition of the DiD Estimator

This appendix formalizes why compositional differences between treatment and control groups can bias the DiD estimator even when subgroup-specific trends are identical across groups.

Consider the simplest 2×2 DiD setup, where superscripts $t$ and $c$ index treatment and control groups and subscripts 0 and 1 index time before and after treatment. The DiD estimator is:

$$\hat{\beta} = (\bar{y}_1^t - \bar{y}_0^t) - (\bar{y}_1^c - \bar{y}_0^c) = \Delta\bar{y}^t - \Delta\bar{y}^c.$$

Now suppose individuals belong to subgroups $s$ with potentially different time trends. Let $\omega_s^i$ denote the share of group $i$ in subgroup $s$, and let $\Delta y_s^i$ denote the change in outcomes for subgroup $s$ within group $i$. The group-level changes can be written as weighted averages:

$$\Delta\bar{y}^t = \sum_s \omega_s^t \Delta y_s^t, \qquad \Delta\bar{y}^c = \sum_s \omega_s^c \Delta y_s^c.$$

Adding and subtracting $\sum_s \omega_s^t \Delta y_s^c$ (the control group's subgroup trends weighted by treatment group composition) yields:

$$\hat{\beta} = \underbrace{\sum_s \omega_s^t \left(\Delta y_s^t - \Delta y_s^c\right)}_{\text{Within-subgroup treatment effect}} + \underbrace{\sum_s \left(\omega_s^t - \omega_s^c\right) \Delta y_s^c}_{\text{Compositional bias}}.$$

The first term is the weighted average of subgroup-specific treatment effects. The second term is the compositional bias: it captures how differences in the distribution of subgroups across treatment and control groups, combined with differential subgroup trends, generate spurious effects.

Under the parallel trends assumption, the second term should be zero. This requires either (1) subgroup trends are identical ($\Delta y_s^c$ is constant across $s$), or (2) the composition of treatment and control groups is balanced ($\omega_s^t = \omega_s^c$ for all $s$).

In the EITC context, neither condition holds. Mothers with two or more children are disproportionately drawn from high-welfare-exposure subgroups, and those subgroups experienced much larger employment increases. As a result, the compositional bias term is large and positive, fully accounting for the observed DiD estimate.

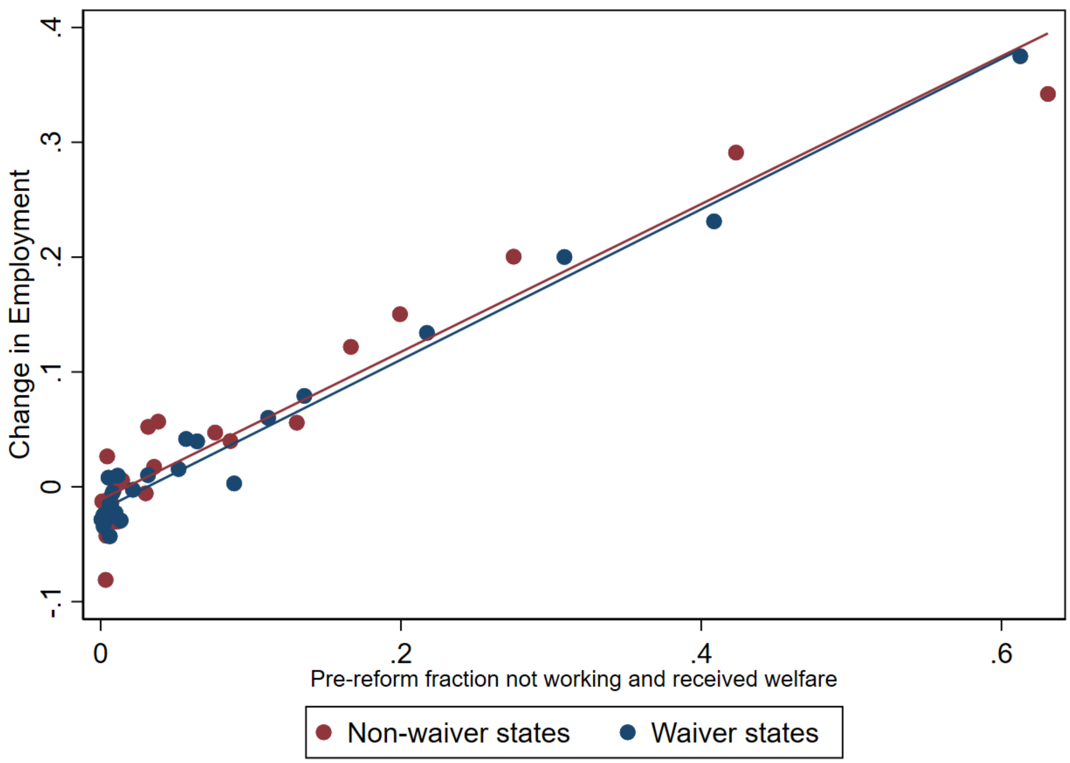
# D Waiver vs. Non-Waiver States

An important robustness check is whether the proportional-exposure pattern varies across waiver and non-waiver states. Schanzenbach and Strain [2021] stress that AFDC waivers generated policy heterogeneity in the early 1990s and argue that including state$\times$year fixed effects accounts for confounding from welfare reform. If the bias I document were solely a product of waiver states, one would expect systematically different relationships in waiver versus non-waiver states.

Figure A2 plots the change in employment from 1993 to 2000 against the pre-reform share of women not working and receiving welfare, separately for waiver and non-waiver states. The slopes are essentially identical, indicating that the proportional-exposure pattern is national, not driven by waiver timing.

This evidence reinforces two points. First, the proportional-exposure mechanism is not confined to waiver states but holds nationally. Second, controlling for state$\times$year effects, as in Schanzenbach and Strain [2021], does not eliminate the bias because the imbalance in exposure occurs within states as well as across them.

Figure A2: Employment Changes by Pre-Reform Welfare Exposure: Waiver vs. Non-Waiver States




Notes: Each point represents a bin of single women grouped by predicted pre-reform welfare exposure, plotted separately for states that implemented AFDC waivers before 1996 (navy) and states that did not (maroon). The horizontal axis shows pre-reform welfare exposure; the vertical axis shows the change in employment between 1993 and 2000. The fitted lines are nearly identical (slopes of 0.62 and 0.67, respectively), indicating that the proportional-exposure pattern is national and not driven by waiver timing. Sample: Single women ages 20–50, March CPS 1991–1993 and 1999–2001.

# E Welfare Exposure: Overlap, EITC Contrast, and Robustness

This appendix presents seven diagnostics that bear on the interpretation of Figures 3 and 4. Appendix Figure A3 documents the overlap in predicted pre-reform welfare exposure $W_i$ between one-child mothers and two-or-more-child mothers. Appendix Table A5 shows that the EITC policy contrast between the two groups remains positive within each welfare-exposure bin, addressing the concern that the reweighting in Figure 3 might mechanically eliminate EITC treatment intensity along with welfare exposure, and Appendix Table A6 shows the same is true for an individual-level simulated EITC evaluated at fixed earnings. Appendix Table A7 reports annual within-bin treatment-control employment gaps with standard errors. Appendix Figure A4 extends the pre-period of Figure 3 back to 1988. Appendix Table A8 shows that the slope underlying Figure 4's placebo bias falls close to zero once Year $\times$ predicted pre-reform welfare exposure is included as a control. Appendix Table A9 shows that, conditional on the welfare-exposure gap, the simulated-EITC contrast has no explanatory power for the placebo estimates.

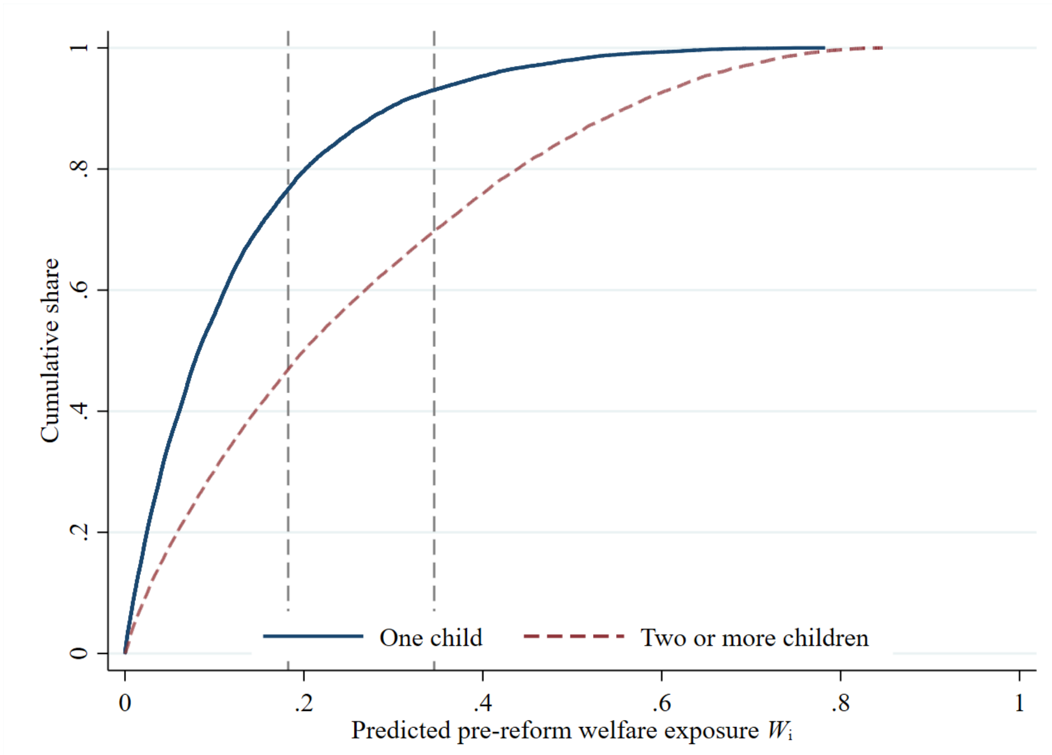
## E.1 Overlap of Welfare Exposure by Family Size

Figure A3 plots the distribution of predicted pre-reform welfare exposure $W_i$ separately by family size. The one-child and two-or-more-child groups overlap throughout the range, but mothers with two or more children are concentrated in the higher-exposure cells—the compositional difference that drives the canonical estimates.

## E.2 EITC Contrast Within Welfare-Exposure Bins

A concern about Figure 3 is that conditioning on $W_i$ may also shrink the EITC treatment contrast between one-child and two-or-more-child mothers. Because the demographic covariates that predict welfare exposure also predict earnings, the family-size groups could

Figure A3: Distribution of Predicted Pre-Reform Welfare Exposure by Number of Children




Notes: Weighted cumulative distribution functions (CDFs) of predicted pre-reform welfare exposure $W_i$, computed separately for single mothers with one child (solid navy) and two or more children (dashed maroon). Vertical dashed lines mark the 60th and 80th weighted percentiles of $W_i$ in the pooled mother sample—the cutoffs used to construct the three bins ("Lowest 60%," "60–80%," "Top 20%") in Figure 3. Both family-size groups have support throughout the range, but mothers with two or more children are substantially more concentrated in high-exposure cells. Sample: single mothers ages 20–50, March CPS 1991–2001; weights are March CPS supplement weights (`asecwt`).

in principle have similar EITC schedules in addition to similar pre-reform welfare exposure within bin.

Appendix Table A5 addresses this concern using the Hoynes–Patel simulated EITC measure. This measure varies at the year × state × number-of-children level, so within a bin it captures the variation in the EITC policy schedule between one-child and two-or-more-child mothers, holding individual demographics fixed. The table shows that the EITC schedule contrast remains positive and substantial in every welfare-exposure bin. The reweighting in Figure 3 therefore does not mechanically eliminate EITC schedule variation between the family-size groups: the within-bin near-zero employment contrasts cannot be attributed to absence of EITC policy contrast.

I interpret this exercise as a diagnostic rather than as a standalone identification strategy. A natural extension is to evaluate the credit at the individual level rather than at the schedule level. Appendix Table A6 constructs a simulated EITC at fixed earnings: the earnings of workers observed in 1998–2001 are expressed in each law year's dollars, run through that year's federal and state EITC schedules using the NBER TAXSIM calculator, and imputed to all sample members from regressions of the simulated credit on the same pre-reform covariates used to construct $W_i$, estimated separately by law year and family-size group. Because earnings and characteristics are held fixed, the measure varies over time only through statutory changes. The individual-level incentive contrast survives within every welfare-exposure bin—the within-bin DD is \$758, \$893, and \$839 (2005 dollars) in the lowest-60%, 60–80%, and top-20% bins. If anything, it is larger than the schedule-level contrast in Appendix Table A5, because the post-reform earnings of single mothers lie near the maximum-credit region of the schedule, where the two-or-more-child expansion was largest. Two caveats apply: post-reform earnings are themselves equilibrium outcomes of the reform period, and late-1990s real wage growth means imputed earnings slightly overstate what the same women would have earned earlier in the decade; both affect the level of the simulated credit more than the family-size contrast used here. The measure has cell-level rather than individual resolution—the covariates explain roughly 15 percent of the individual variation in the simulated credit among mothers—which is the relevant resolution for the within-bin and placebo-pair comparisons that use it.

Table A5: EITC Policy Contrast Within Welfare-Exposure Bins

| | Pre (1991–1993) | | Post (1999–2001) | | Within-bin |
|---|---|---|---|---|---|
| Welfare-exposure bin | One child | Two or more | One child | Two or more | DD |
| Lowest 60% | $ 685 | $ 703 | $ 923 | $1,450 | $ 509 |
| 60–80% | $ 686 | $ 696 | $ 938 | $1,452 | $ 505 |
| Top 20% | $ 679 | $ 687 | $ 957 | $1,456 | $ 491 |

Notes: Weighted mean of the Hoynes–Patel simulated EITC (2005 dollars) by welfare-exposure bin, family size, and period. The simulated EITC varies at the year × state × number-of-children level—not at the individual-earnings level—so the table captures the EITC schedule contrast between one-child and two-or-more-child mothers within each welfare-exposure bin, holding individual demographics fixed within bin. The within-bin DD in the final column equals [(2+ post − 2+ pre) − (1-child post − 1-child pre)]. Welfare-exposure bins are defined by the same 60th and 80th weighted percentiles of pre-reform welfare exposure $W_i$ used in Figure 3. Sample: single mothers ages 20–50, March CPS 1991–1993 and 1999–2001; weights are March CPS supplement weights (`asecwt`).

Table A6: Simulated EITC at Fixed Earnings Within Welfare-Exposure Bins

| | Pre (1991–1993) | | Post (1999–2001) | | Within-bin |
|---|---|---|---|---|---|
| Welfare-exposure bin | One child | Two or more | One child | Two or more | DD |
| Lowest 60% | $ 790 | $ 800 | $1,163 | $1,931 | $ 758 |
| 60–80% | $1,057 | $1,069 | $1,643 | $2,548 | $ 893 |
| Top 20% | $1,118 | $1,115 | $1,821 | $2,657 | $ 839 |

Notes: Weighted mean of an individual-level simulated EITC (federal plus state, 2005 dollars) by welfare-exposure bin, family size, and period. The simulated EITC is constructed by fixing the earnings of workers observed in 1998–2001, expressing them in each law year's dollars, computing the federal and state EITC under each law year 1991–2001 using the NBER TAXSIM calculator, and imputing the result to all sample members from regressions of the simulated credit on the same pre-reform covariates used to construct welfare exposure $W_i$, estimated separately by law year and family-size group. Because earnings and characteristics are held fixed, variation over time reflects only statutory changes in the credit. The within-bin DD in the final column equals [(2+ post − 2+ pre) − (1-child post − 1-child pre)]. Welfare-exposure bins are defined by the same 60th and 80th weighted percentiles of pre-reform welfare exposure $W_i$ used in Figure 3. Sample: single mothers ages 20–50, March CPS 1991–1993 and 1999–2001; weights are March CPS supplement weights (`asecwt`).

## E.3 Annual Within-Bin Gaps With Standard Errors

Figure 3 plots annual mean employment by family size and welfare-exposure bin without standard errors. Appendix Table A7 reports the underlying annual means, the standard error of each, the treatment-control gap (two-or-more-child minus one-child) within each bin, and its standard error (treating the two subsamples as independent). The high-exposure (Top 20%) bin has the smallest one-child cell, as expected from the composition statistics in Section 4.1, but the gap and its standard error are small in magnitude through the pre-period, supporting the interpretation that the within-bin trajectories are nearly identical rather than only superficially so. In the post-period the top-bin gap averages five to six percentage points, with annual standard errors of three to seven points. Part of this is the exposure gradient operating within the bin: two-or-more-child mothers have higher welfare exposure even within the top bin, with a pre-period welfare-receipt gap of 0.055 that puts 1.4 percentage points on the placebo line of Figure 4. The remainder is within the sampling variability of the smallest cell, which includes the outlier 1993 one-child observation (employment of 0.482 on 175 observations, against roughly 0.38–0.41 in the preceding years).

## E.4 Extended Pre-Period

A separate concern is that the high-exposure cells could have been temporarily depressed in 1991–1993, in which case part of the post-1993 within-bin employment increase might reflect mean reversion rather than the underlying trend. Appendix Figure A4 extends the pre-period of Figure 3 back to 1988, using the same propensity-score model (estimated on 1991–1993 data) and the same bin cutoffs. Within-bin employment levels in 1988–1990 are similar to those in 1991–1993, indicating that the 1991–1993 high-exposure cells were not unusually depressed in the immediate pre-reform years.

## E.5 Placebo Bias and Year $\times$ Welfare Exposure

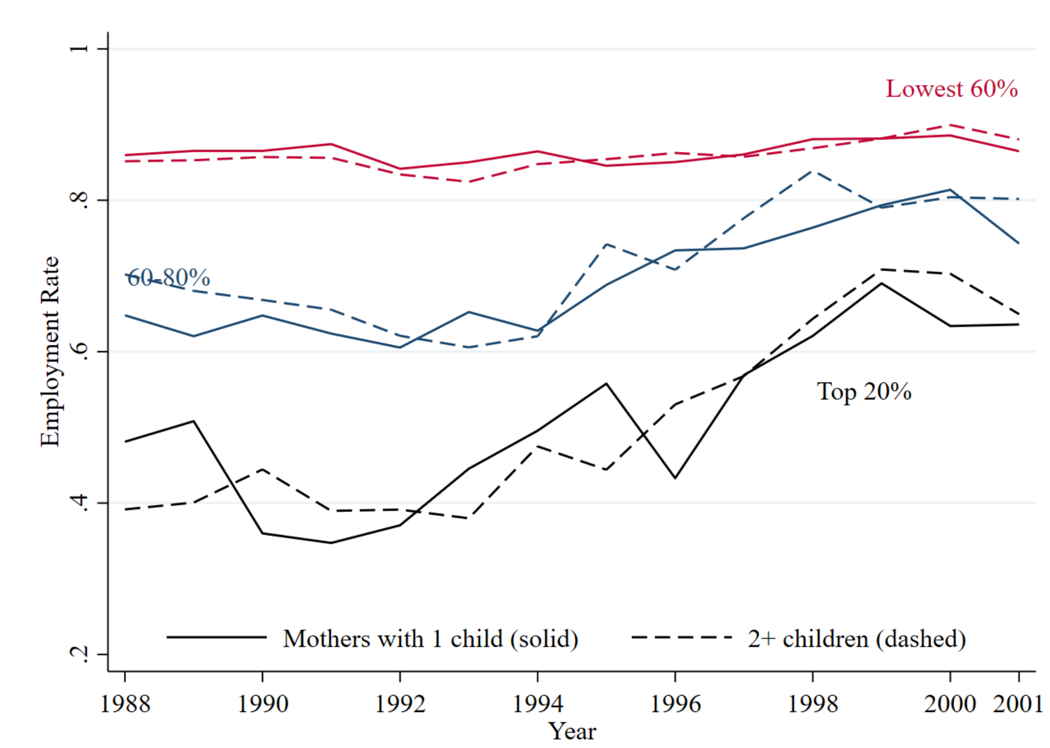
The omitted-variable interpretation of Figure 4 has a sharp testable implication: the same variable that predicts the placebo bias should eliminate it once included as a control. Appendix Table A8 reports this exercise. In the within-EITC-group placebo comparisons used to construct Figure 4, the slope of the placebo coefficient on the pre-reform welfare-exposure

Table A7: Annual Within-Bin Treatment–Control Employment Gaps with Standard Errors

| | One child | | | Two or more | | | Gap (2+ − 1ch) | |
|---|---|---|---|---|---|---|---|---|
| Year | Mean | SE | N | Mean | SE | N | Gap | SE |
| *Panel A: Lowest 60%* | | | | | | | | |
| 1991 | 0.872 | (0.009) | 1,528 | 0.852 | (0.012) | 1,281 | -0.020 | (0.015) |
| 1992 | 0.843 | (0.008) | 1,555 | 0.838 | (0.008) | 1,374 | -0.005 | (0.012) |
| 1993 | 0.851 | (0.013) | 1,663 | 0.833 | (0.010) | 1,332 | -0.018 | (0.016) |
| 1994 | 0.858 | (0.015) | 1,661 | 0.849 | (0.012) | 1,360 | -0.009 | (0.020) |
| 1995 | 0.848 | (0.011) | 1,509 | 0.857 | (0.012) | 1,265 | 0.010 | (0.016) |
| 1996 | 0.853 | (0.012) | 1,552 | 0.861 | (0.013) | 1,276 | 0.008 | (0.018) |
| 1997 | 0.868 | (0.008) | 1,591 | 0.857 | (0.010) | 1,265 | -0.011 | (0.012) |
| 1998 | 0.881 | (0.010) | 1,527 | 0.867 | (0.011) | 1,285 | -0.014 | (0.015) |
| 1999 | 0.880 | (0.009) | 1,532 | 0.884 | (0.012) | 1,249 | 0.004 | (0.015) |
| 2000 | 0.886 | (0.008) | 2,828 | 0.895 | (0.008) | 2,398 | 0.009 | (0.012) |
| 2001 | 0.867 | (0.007) | 2,908 | 0.881 | (0.010) | 2,480 | 0.014 | (0.012) |
| *Panel B: 60–80%* | | | | | | | | |
| 1991 | 0.596 | (0.028) | 396 | 0.671 | (0.016) | 664 | 0.075 | (0.032) |
| 1992 | 0.597 | (0.035) | 370 | 0.619 | (0.027) | 623 | 0.022 | (0.044) |
| 1993 | 0.631 | (0.021) | 408 | 0.597 | (0.023) | 624 | -0.034 | (0.031) |
| 1994 | 0.658 | (0.028) | 374 | 0.633 | (0.024) | 654 | -0.024 | (0.037) |
| 1995 | 0.675 | (0.028) | 342 | 0.727 | (0.028) | 560 | 0.053 | (0.040) |
| 1996 | 0.726 | (0.028) | 376 | 0.719 | (0.023) | 607 | -0.006 | (0.037) |
| 1997 | 0.707 | (0.029) | 320 | 0.776 | (0.019) | 596 | 0.069 | (0.035) |
| 1998 | 0.756 | (0.019) | 349 | 0.830 | (0.017) | 558 | 0.074 | (0.026) |
| 1999 | 0.810 | (0.024) | 356 | 0.780 | (0.017) | 634 | -0.030 | (0.029) |
| 2000 | 0.808 | (0.021) | 650 | 0.816 | (0.012) | 1,038 | 0.008 | (0.025) |
| 2001 | 0.749 | (0.028) | 664 | 0.798 | (0.014) | 1,050 | 0.049 | (0.031) |
| *Panel C: Top 20%* | | | | | | | | |
| 1991 | 0.405 | (0.045) | 199 | 0.396 | (0.022) | 967 | -0.009 | (0.050) |
| 1992 | 0.381 | (0.049) | 180 | 0.396 | (0.017) | 907 | 0.015 | (0.052) |
| 1993 | 0.482 | (0.047) | 175 | 0.384 | (0.018) | 964 | -0.098 | (0.050) |
| 1994 | 0.515 | (0.035) | 195 | 0.479 | (0.022) | 938 | -0.036 | (0.041) |
| 1995 | 0.562 | (0.047) | 148 | 0.463 | (0.027) | 845 | -0.098 | (0.054) |
| 1996 | 0.467 | (0.026) | 161 | 0.532 | (0.027) | 793 | 0.065 | (0.038) |
| 1997 | 0.566 | (0.066) | 134 | 0.573 | (0.029) | 759 | 0.007 | (0.072) |
| 1998 | 0.655 | (0.051) | 131 | 0.657 | (0.035) | 790 | 0.002 | (0.062) |
| 1999 | 0.666 | (0.055) | 148 | 0.720 | (0.027) | 737 | 0.053 | (0.062) |
| 2000 | 0.650 | (0.029) | 248 | 0.706 | (0.017) | 1,220 | 0.055 | (0.034) |
| 2001 | 0.600 | (0.039) | 263 | 0.662 | (0.017) | 1,175 | 0.062 | (0.043) |

Notes: Annual mean employment (`earnany`) for single mothers by welfare-exposure bin and family size. Standard errors are clustered at the state level. The gap is the difference in mean employment between two-or-more-child mothers and one-child mothers within the same welfare-exposure bin in the same year; its standard error treats the two subsamples as independent. $N$ is the unweighted cell sample size. Companion to the top panel of Figure 3. Sample: single mothers ages 20–50, March CPS 1991–2001; weights are March CPS supplement weights (`asecwt`).

Figure A4: Employment Rates of Single Mothers by Predicted Pre-Reform Welfare Use, Extended Pre-Period (1988–2001)




Notes: Extended pre-period analog of the top panel of Figure 3. Welfare-exposure groups, family-size groups, the propensity-score model, and the bin cutoffs are identical to Figure 3; the only change is extending the sample from 1991–2001 to 1988–2001. Within-bin employment levels in 1988–1990 are similar to those in 1991–1993, which is inconsistent with a temporary pre-1991 depression in the high-exposure cells as the source of the subsequent gains. Sample: single mothers ages 20–50, March CPS 1988–2001.

gap is large in the baseline specification. Adding Year × predicted pre-reform welfare exposure as a control drives this slope close to zero, confirming that the placebo bias loads on differential exposure to welfare reform.

Table A8: Placebo Bias With and Without Year × Pre-Reform Welfare Exposure

| Specification | Slope on welfare-exposure gap | SE | $R^2$ | N |
|---|---|---|---|---|
| Baseline placebo DiD | 0.361 | (0.007) | 0.850 | 540 |
| + Year × $W_i$ | 0.007 | (0.016) | 0.000 | 540 |

Notes: Each row reports the OLS slope of the placebo DiD coefficient on the pre-reform welfare-exposure gap between placebo "treatment" and "control" groups, fit over the within-EITC-group placebo estimates with $|\Delta W| < 0.9$ and $\Delta W \geq 0$ (matching Figure 4). The baseline specification reproduces Figure 4's fitted slope. The second row adds Year × predicted pre-reform welfare exposure $W_i$ to each placebo regression. The variable that predicts placebo bias eliminates it once included as a control. Standard errors treat the placebo pairs as independent; because pairs share decile cells, they overstate precision, and the table is descriptive. Sample for the placebo regressions: single women ages 20–50, March CPS 1991–2001; weights are March CPS supplement weights (`asecwt`). Each underlying placebo regression absorbs state-by-year fixed effects.

## E.6 Placebo Estimates: Welfare Exposure vs. EITC Incentives

A final diagnostic asks which variable organizes the placebo estimates of Figure 4: the welfare-exposure gap between the placebo cells, or the contrast in EITC incentives. For every within-group placebo pair, I compute the pre-reform welfare gap $\Delta W$ and the contrast $\Delta$EITC in the individual-level simulated EITC of Appendix Table A6—the change in the simulated credit between 1991–1993 law and 1994–2001 law in the placebo "treatment" cell minus the same change in the "control" cell, matching the pre and post windows of the placebo regressions. Because the simulated credit holds earnings and characteristics fixed, $\Delta$EITC isolates statutory incentive variation across cells. Appendix Table A9 regresses the placebo estimates on each variable separately and jointly. Alone, $\Delta$EITC appears to predict the placebo estimates ($R^2 = 0.40$), but only because it is correlated with welfare exposure (the pair-level correlation between $\Delta W$ and $\Delta$EITC is 0.72). Conditional on $\Delta W$, the $\Delta$EITC coefficient is slightly negative and economically negligible, explained variation is unchanged ($R^2$ rises from 0.850 to 0.852), and the coefficient on $\Delta W$ is unaffected. The same pattern holds within the two-or-more-child group, where incentive variation across cells is largest, and within the predicted-earnings dimension, where cells differ most sharply in ΔEITC. The placebo estimates are organized by welfare exposure, not by EITC incentives.

Table A9: What Predicts the Placebo Estimates: Welfare Exposure vs. EITC Incentives

| | (1) | (2) | (3) |
|---|---|---|---|
| Pre-reform welfare gap ($\Delta W$) | 0.3607 | | 0.3793 |
| | (0.0069) | | (0.0109) |
| Simulated-EITC contrast (ΔEITC, thousands of 2005 dollars) | | 0.2603 | -0.0273 |
| | | (0.0165) | (0.0109) |
| $R^2$ | 0.850 | 0.397 | 0.852 |
| Placebo pairs | 540 | 540 | 540 |

Notes: Each column is an OLS regression across within-group placebo pairs; the dependent variable is the placebo DiD estimate. Placebo pairs assign artificial treatment and control labels to decile cells of pre-reform propensity scores (welfare receipt, employment, welfare-and-not-working, and predicted earnings) within groups facing the same EITC schedule (childless women, one-child mothers, two-or-more-child mothers), exactly as in Figure 4. $\Delta W$ is the pre-reform (1991–1993) difference in welfare receipt between the artificial treatment and control cells. ΔEITC is the difference between the cells in the change in the individual-level simulated EITC (Appendix Table A6) from 1991–1993 to 1994–2001, in thousands of 2005 dollars; because the simulated EITC holds earnings and characteristics fixed, this contrast reflects only statutory changes in the credit. The sample is restricted to pairs with $0 \leq \Delta W < 0.9$, as in Figure 4. Robust standard errors in parentheses; they treat the pairs as independent, and because pairs share decile cells they overstate precision—the table is descriptive.

# F Back-of-the-Envelope Simulation

The caseload reduction hypothesis makes specific empirical predictions about which mothers would be affected and by how much. If the policy operated by reducing the likelihood of successfully applying for or renewing welfare benefits and was implemented at the caseworker-recipient level, its effects should be proportional to the rate of welfare use before the reform and the gap between the employment of single mothers and comparable single women. However, because there were large differences in the propensity to use welfare in the early 1990s, an equal proportional decline in welfare use (and increase in employment) would result in large differences in levels (measured, for example, as percentage-point changes).

Before reform, for example, 32 percent of single mothers reported receiving cash welfare. After reform, the rate was 11 percent, a 65 percent (or 21 percentage point) decline. The decline in welfare among mothers of one child was 16 percentage points (from 24 percent to 8 percent), while the decline among mothers with two or more children was 24 percentage points (from 38 percent to 14 percent). Likewise, the rate of welfare use of mothers with children less than 3 fell by 34 percentage points, whereas it fell by only 6 percentage points among mothers whose youngest child was 13 or older.

## F.1 Simulation

To illustrate, I simulate a policy intervention that removes single mothers from welfare at random and in numbers sufficient to achieve annual caseload reductions observed in the March CPS. Nonworking mothers are assigned to find work at random. By design, the "effect" of the simulated policy is unconditional on the characteristics of the mother (such as level of education, number of children, or age of youngest child). Implemented in the March CPS, I compare the effect of this placebo policy, both qualitatively and econometrically, to the evidence of the EITC's effect, including the difference-in-difference estimators widely used in this literature.

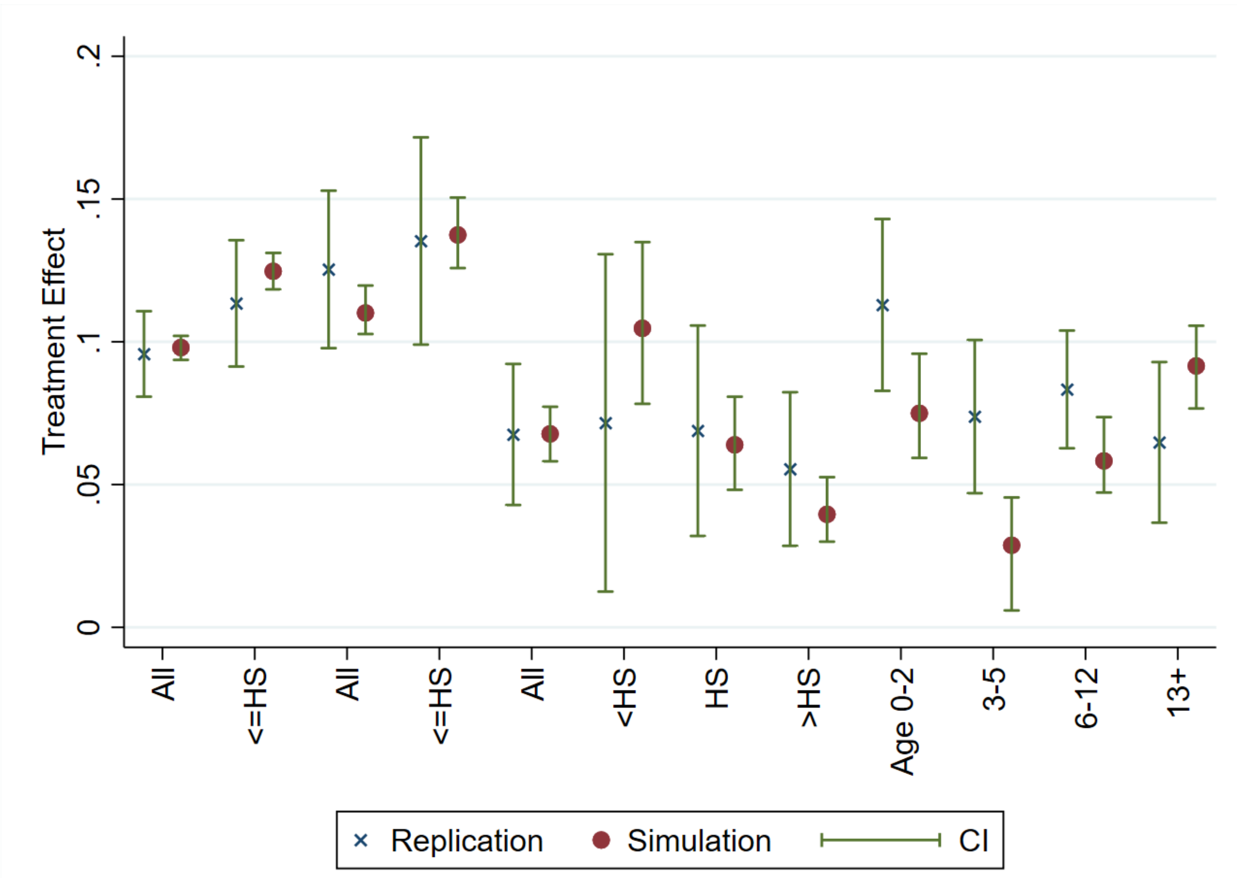
The simulation procedure is as follows: I construct a synthetic panel data set of annual micro data from the March CPS. I start with a pre-reform March CPS cross section (e.g. data from 1993) of single mothers and childless women (identical to that used in the analysis above), stack eleven of those cross sections together, and label each cross section with a different year from 1991 to 2001. Then, for each of those "synthetic" years, I randomly assign a fraction of welfare recipients to be "kicked off" in proportion to the amount needed to achieve the actual fraction of individuals on welfare in each year from 1991 through 2001. For example, in 1993 33 percent of single mothers received welfare in the March CPS. In 2000, only 10 percent received welfare, a 68 percent reduction. Hence, for the synthetic year 2000 cross section, I randomly assign 68 percent of observed welfare recipients to not be on welfare (i.e. the dummy variable is recoded from 1 to 0 in 68 percent of cases). Likewise, if the employment rate rises from 70 percent to 80 percent between 1993 and a subsequent year, then I assume the job finding was at random: I assign 33 percent of nonworking mothers ((80-70)/(100-70)) to be employed. The result of the exercise is an 11-year panel dataset with the same covariates as are available in the March CPS, but in which the only time series variation is the result of randomly assigning observations to not be on welfare and work.

I estimate the same regression specifications as above. The results, presented in Figure A5, yield two findings.

First, the TWFE coefficient estimates from the simulated data closely match those from the actual analysis, despite the fact that simulated welfare exits and job entries are assigned entirely at random with no differential treatment by number of children, education, or any other characteristic. The implication is that an equal proportional reduction in welfare use—consistent with the administrative changes and work requirements described in the ethnographic literature—mechanically generates the same DiD estimates as those reported in the canonical studies. Put differently, the regression specifications cannot distinguish between the EITC and a simple, uniform caseload reduction.

Second, the simulation rationalizes several "puzzles" in the data that have been cited as

Figure A5: Simulation vs. Replication: Estimated Effect of EITC

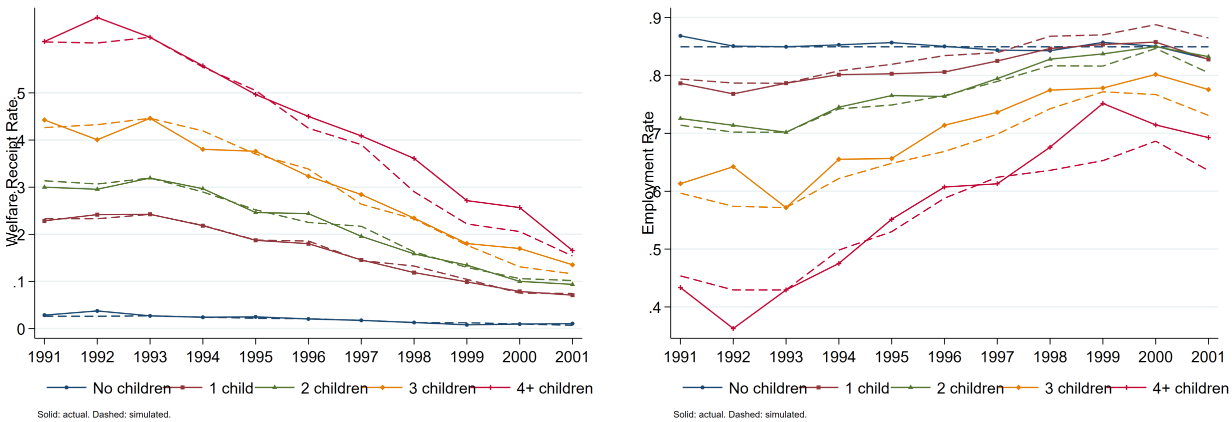



Notes: "×" markers show EITC coefficient estimates from baseline replications in the March CPS. Circles show estimates from simulated data in which the only time-series variation comes from randomly removing mothers from welfare rolls and assigning them to employment at rates matching aggregate CPS trends—with no differential treatment by number of children or other characteristics. The simulation reproduces the canonical DiD estimates, demonstrating that an equal proportional reduction in welfare use (consistent with the caseload reduction hypothesis) mechanically generates the observed treatment effects. Vertical bars are 95% confidence intervals for the replication estimates and the range of estimates across the 300 Monte Carlo draws for the simulation estimates. Sample: Single women ages 20–50, March CPS 1991–2001.

evidence for or against the EITC interpretation. Kleven [2024], for example, highlights the fanning-out of employment gains by family size and the disproportionately large responses among groups with the highest pre-reform welfare exposure as anomalies that are difficult to reconcile with EITC incentives alone. Figures A6 and A7 show that the random-exit simulation accurately predicts these patterns. Figure A6 plots welfare receipt and employment by number of children (0 through 4+), comparing actual trends (solid lines) to simulated trends (dashed lines). The simulation reproduces both the sharp decline in welfare use and the corresponding increase in employment across all groups, including the pronounced fanning-out among mothers with three or four or more children. Figure A7 presents the same comparison by age of youngest child, using the categories from Michelmore and Pilkauskas [2021] (0–2, 3–5, 6–12, 13+). Here too the simulated trends closely track the actual data: the largest declines in welfare and the largest increases in employment occur among mothers with the youngest children—precisely the groups with the highest pre-reform welfare exposure—and the simulation matches these differential changes without any heterogeneous treatment.

The close correspondence between simulated and actual outcomes across both dimensions corroborates the conclusion that the employment gains of the 1990s are fully consistent with a uniform, proportional reduction in welfare caseloads. Because the simulation contains no EITC variation whatsoever, the fact that it replicates both the regression estimates and the subgroup-level trends carries no information that distinguishes an EITC response from a uniform caseload reduction.

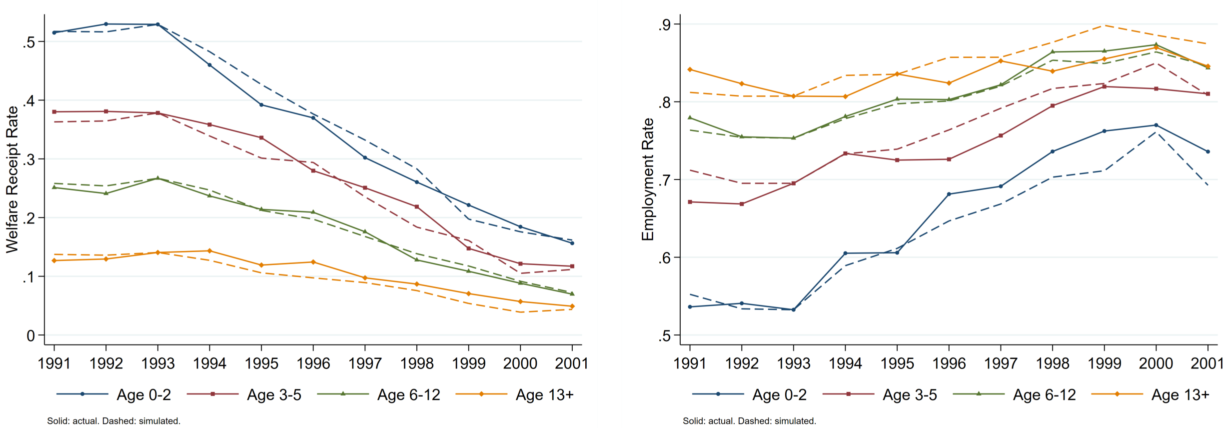
Figure A6: Actual vs. Simulated Welfare Receipt and Employment by Number of Children




Welfare Receipt Employment

Notes: Solid lines with markers show actual annual means from the March CPS; dashed lines show means from one draw of the simulation in which welfare exits and job entries are assigned uniformly at random to match aggregate CPS trends. No differential treatment is applied by number of children. The simulation reproduces the fanning-out pattern across family sizes. Sample: Single women ages 20–50, March CPS 1991–2001.

Figure A7: Actual vs. Simulated Welfare Receipt and Employment by Age of Youngest Child




Welfare Receipt Employment

Notes: Solid lines with markers show actual annual means from the March CPS; dashed lines show simulated means. Age-of-youngest-child categories follow Michelmore and Pilkauskas [2021]: 0–2, 3–5, 6–12, and 13+. Mothers only (restricted to women with at least one child). The simulation reproduces the heterogeneous changes by age of youngest child without any differential treatment. Sample: Single mothers ages 20–50, March CPS 1991–2001.

# G Time Limits

In a related literature, Grogger [2004] finds that welfare reform's newly required five-year time limit on cumulative welfare use was a significant contributor to the decline in welfare participation and increase in employment during the late 1990s.

This literature is relevant to the concern that the EITC is confounded by the effects of welfare reform because the evidence on the effects of welfare time limits uses a nearly identical identification strategy and regression specification, but very different treatment and control groups based on the age of children (but not the number of children). As a result, if welfare reform is a confounding factor in the analysis of the EITC it is also likely to have the same confounding effect on estimates of welfare time limits. Moreover, finding a confounding effect of welfare reform on time limits corroborates and reinforces the analysis above.

In theory, these time limits generate an immediate incentive to reduce welfare use to conserve benefits as insurance against future economic shocks. However, this precautionary motive should only affect families with younger children because the typical five-year time limit is not binding on families where the youngest child is aged 13 to 17; their eligibility already will end when the child turns 18 before the limit could be reached. Grogger shows that the level of welfare participation of families with young children fell faster than that of families with older children after the implementation of time limits and interprets this as evidence of an anticipatory response to time limits.

The econometric analysis of the effect of welfare time limits follows the same DiD specification as the EITC with an additional set of policy variables for the impact of time limits. In particular, in Grogger's 2004 specification, the effect of time limits is captured with a dummy variable for whether the state has a welfare time limit in place in that year interacted with two age-of-youngest child dummy variables (for "0–6" and "7 to 12") plus dummy variables for the age of youngest child.

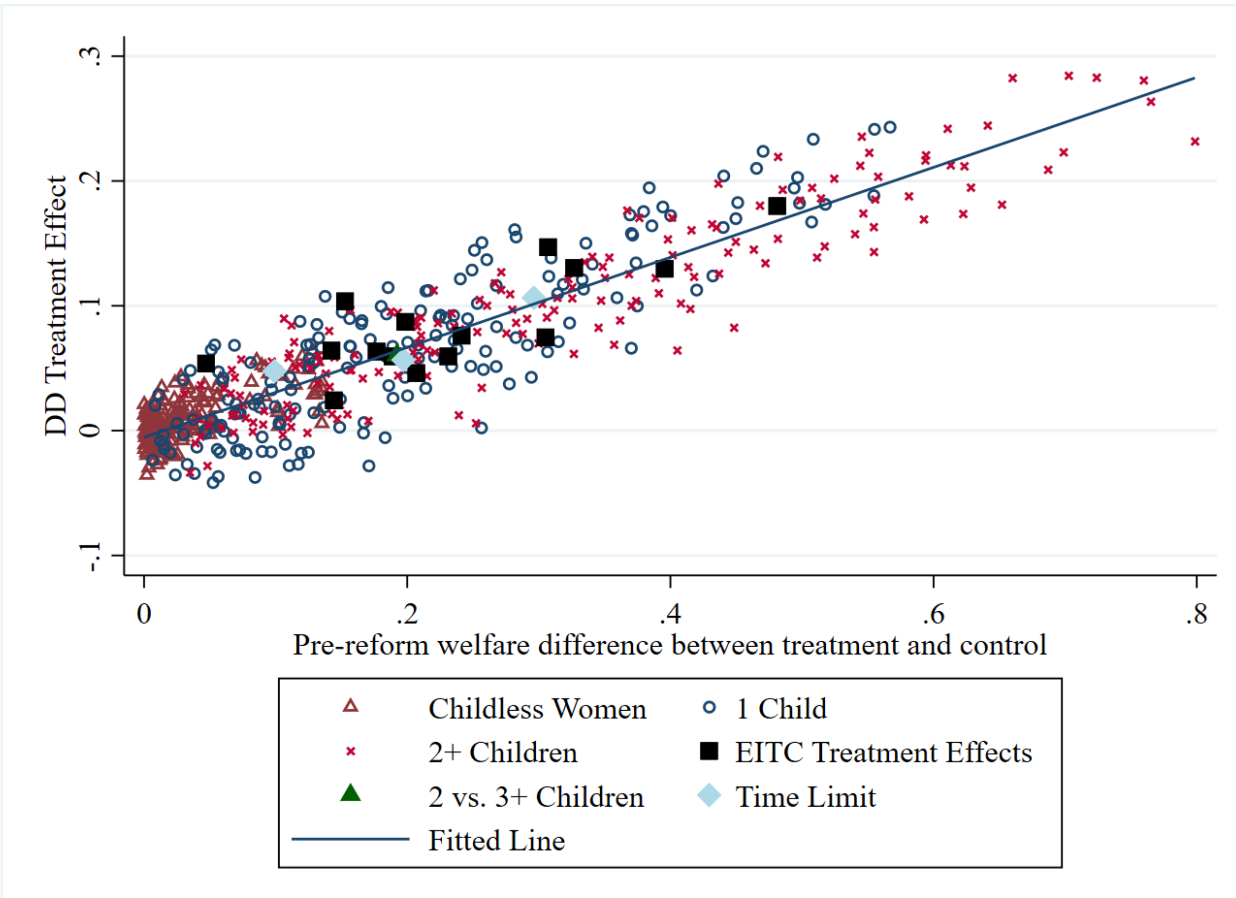
## G.1 Analysis

In my reanalysis of time limits, I replicate the same general approach as I take to the EITC. First I reproduce the analysis in a consistent sample from 1991–2001 and compare it to published estimates, corroborating the original analysis. In my replication, I exclude mothers whose oldest child is 18, include the maximum value of the EITC in each year for each group (single women, mothers with one child, mothers of 2+ children). This approach uses the simplest specification from Grogger [2004] for clarity, though the results are the same applying the exact specifications.

Second, I compare the DiD coefficient estimates to placebo estimates and show that these estimates are similarly predictable based on the ex-ante differences in welfare use between treatment and control observations.

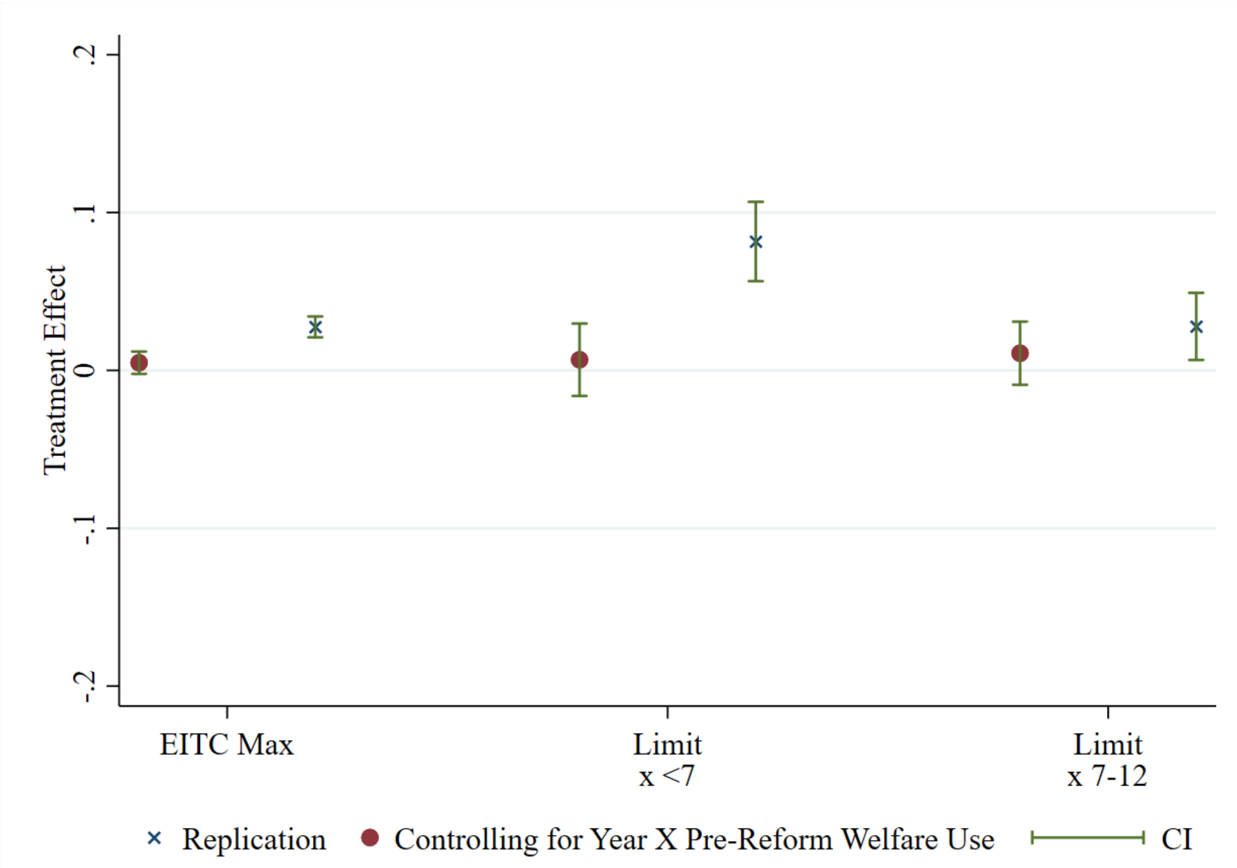
Finally, I include controls for pre-reform welfare exposure by time effects—the same Year $\times$ predicted pre-reform welfare exposure interactions used in Section 3—and show that the estimated effect of time limits on employment is attenuated toward zero, just as the EITC effects were.

Figure A8: Placebo and Actual DiD Estimates: EITC and Welfare Time Limits




Notes: Hollow symbols show placebo DiD estimates as in Figure 4. Black squares are actual EITC treatment effect estimates from pairwise comparisons in the published literature. The solid green triangle compares mothers of three or more children to mothers of exactly two, who faced an identical EITC schedule after 1993. Blue diamonds are DiD estimates of the effect of welfare time limits following Grogger [2004], comparing mothers whose youngest child is under 13 (affected by time limits) to mothers whose youngest child is 13–17 (unaffected because welfare eligibility ends at 18). Time limit estimates fall on the same fitted line as EITC estimates, indicating that both reflect the same underlying confound: differential exposure to welfare reform. Sample: Single women ages 20–50, March CPS 1991–2001.

Figure A9: Time Limit Estimates With and Without Welfare Exposure Controls




Notes: Estimates of the effects of the EITC ("EITC Max") and welfare time limits ("Limit × <7" and "Limit × 7–12") on employment, following Grogger [2004]. "×" markers show baseline replication estimates; circles show estimates after adding Year × predicted pre-reform welfare exposure interactions. Vertical bars are 95% confidence intervals. As with the EITC specifications, controlling for welfare exposure attenuates the time limit coefficients toward zero. Sample: Single mothers ages 20–50 with youngest child under 18, March CPS 1991–2001.